Capstone Project: Bellabeat Wellness Case Study

[How can a Wellness Company Play it Smart?]

Executive Summary

Bellabeat, a wellness technology company, sought to understand how consumers use smart devices to inform marketing strategy for its product line. Using Fitbit data from 30 users, I built a complete analytical pipeline—from raw data ingestion to semantic-layer modeling and dashboard development—to uncover behavioral patterns in daily activity, sleep, hourly movement, minute-level intensity, and overall wellness.

Across all M3 semantic-layer tables, the analysis revealed several consistent trends. Users are moderately active but rarely engage in high-intensity exercise, sleep less than recommended, and show significant drops in activity on weekends. Sleep efficiency varies widely, and users often spend more time in bed than asleep. Strong correlations exist between steps and calories burned, while sleep and activity show weak alignment, reinforcing the need for holistic wellness guidance.

These insights informed three strategic recommendations for Bellabeat:

1. Promote habit-building features to increase daily activity and reduce weekend inactivity.
2. Highlight sleep-tracking and wellness insights to address insufficient sleep patterns.
3. Position Bellabeat as a holistic wellness ecosystem that integrates activity, sleep, stress, hydration, and women's health.

This case study demonstrates how raw sensor data can be transformed into actionable business intelligence through structured data engineering, SQL analysis, and interactive dashboards. The findings provide Bellabeat with clear, data-driven opportunities to strengthen product positioning, enhance user engagement, and support long-term wellness outcomes.

ASK Phase – Define the Business Task

Business Problem: Bellabeat wants to understand how consumers use smart wellness devices, particularly non-Bellabeat products such as Fitbit trackers. By analyzing real user behavior, the company aims to identify trends that can guide marketing strategy for one of its products.

Purpose of the Analysis:

The insights from this project will help Bellabeat:

- Understand daily habits related to activity, sleep, steps, and heart rate.
- Identify behavioral trends that influence wellness product usage.
- Apply these trends to Bellabeat's product line (Leaf, Time, Spring, Bellabeat App).
- Shape targeted marketing strategies based on real user behavior.
- Support decisions about product positioning and customer engagement.

Stakeholders

- Urška Sršen — Cofounder & Chief Creative Officer.
- Sando Mur — Cofounder.
- Bellabeat Marketing Analytics Team.
- Bellabeat Executive Team.

Guiding Questions

- What are some trends in smart device usage?
- How could these trends apply to Bellabeat customers?
- How could these trends help influence Bellabeat marketing strategy?

ASK Stage – Final Deliverable

Business Task Statement: Using the Fitbit Fitness Tracker dataset, I will analyze user activity, sleep behavior, hourly movement, minute-level intensity, and overall wellness patterns to identify key trends in smart-device usage. These insights will be applied to one Bellabeat product to develop high-level marketing recommendations that support Bellabeat's growth strategy and improve customer engagement.

PREPARE — Collect & Understand the Data

Data Source

The dataset used in this case study is the FitBit Fitness Tracker Data available on Kaggle under a CC0 Public Domain license. It contains activity, sleep, steps, intensities, heart-rate, and weight-log data collected from 30 Fitbit users over a two-month period. Because the data is publicly available and anonymized, it is safe to use and free of licensing restrictions.

Data Organization

The dataset is structured into multiple CSV files, each representing a different category of Fitbit data:

- dailyActivity.
- sleepDay.
- dailySteps.
- dailyIntensities.
- heartrate_seconds.
- weightLogInfo.

Each file contains timestamped records linked by ***a unique Id,*** allowing the datasets to be joined for deeper analysis.

ROCCC Assessment (Credibility Check)

Using the ROCCC framework:

- **Reliable:** Partially — small sample size (30 users).
- **Original:** Yes — collected directly from Fitbit devices.
- **Comprehensive:** Limited — no demographic information.
- **Current:** No — data is from 2016.
- **Cited:** Yes — properly sourced and licensed.

Conclusion: The dataset is useful for identifying general behavioral trends but has limitations that require cautious interpretation.

Licensing, Privacy & Accessibility

- CC0 license → free to use without restrictions.
- No personally identifiable information (PII).
- All files stored securely with original copies preserved.
- Accessible through spreadsheets, SQL, and Python.

Data Integrity Verification

To ensure the dataset was suitable for analysis, I:

- Confirmed all CSV files loaded correctly.
- Checked for consistent column names and formats.
- Verified timestamp accuracy.
- Identified and removed duplicates (especially in sleep logs).
- Checked for missing values and inconsistencies.
- Ensured user IDs aligned across datasets.

Why Does This Data Support the Business Task?

The Fitbit dataset provides detailed information on:

- Daily activity levels.
- Steps.
- Calories burned.
- Sleep duration and quality.
- Heart-rate patterns.
- Weight logs.

These metrics directly support Bellabeat's business questions by revealing:

- User habits.
- Activity consistency.
- Sleep behavior.
- Wellness routines.

This makes the dataset suitable for identifying smart-device usage trends that can inform Bellabeat's marketing strategy.

Data Limitations

- Small sample size.
- Outdated (2016).
- Missing sleep records.
- Sparse weight and heart-rate data.
- Duplicates in sleep logs.
- Highly granular HR data requiring heavy cleaning.

These limitations mean insights should be interpreted as directional rather than definitive.

PREPARE Stage — Final Deliverable

For this analysis, I used the Fitbit Fitness Tracker dataset, a public-domain collection of CSV files containing activity, sleep, and heart-rate data from 30 users. Although the dataset is well-structured, it has limitations such as small sample size, missing demographic information, and outdated timestamps. Despite limitations, the dataset provides valuable insight into smart-device usage patterns that can inform Bellabeat's marketing strategy.

PROCESS — Clean & Prepare the Data

Tools Used

I used **BigQuery SQL** to clean and prepare the Fitbit dataset because it supports fast querying, efficient handling of multiple CSV files, structured data-cleaning workflows, and seamless integration with dashboarding tools. BigQuery also allows repeatable transformations, ensuring consistency across all stages of the pipeline.

M1 → M2 → M3 Data Engineering Pipeline

M1 — Raw Layer

All original Fitbit CSV files were uploaded into BigQuery exactly as provided by Fitabase.

This layer preserved the original schemas, formats, and naming conventions, serving as the immutable source of truth.

Raw tables included: Daily_activity, sleep_day, daily_steps, daily_intensities, heartrate_seconds, weight_log_info, hourly tables, minute-level narrow tables, and minute-level sleep.

M2 — Clean Layer

The M2 layer contains all cleaned and standardized datasets.

Cleaning steps included:

- Removing duplicates **(especially in sleep logs).**
- Converting text-based dates into proper **DATE/DATETIME** formats.
- Standardizing column names to **snake_case.**
- Checking for missing values and sparse fields.
- Verifying user ID consistency across tables.
- Merging **daily, hourly, minute-level, sleep, and weight** tables into unified tables.

This layer ensures the data is accurate, consistent, and ready for analysis.

M3 — Semantic Layer

The M3 layer is the final analytical model used for dashboards and insights.

It includes purpose-built tables optimized for trend analysis, correlations, and visualization:

- **M3_daily_master**— daily steps, distance, calories, activity minutes.
- **M3_hourly_master** — hourly steps, calories, intensity, HR.
- **M3_minuteWide_long** — wide-format minute-level activity.
- **M3_minute_master** — long-format minute-level steps, calories, intensity, HR.
- **M3_sleep_master** — sleep duration, efficiency, stages.
- **M3_user_summary** — user profile metrics.

These tables form the foundation of all dashboards created in the SHARE stage.

Structured Cleaning Workflow (BigQuery SQL)

1. **Uploading Raw Data:** All Fitbit CSVs were imported into BigQuery as separate tables with correct data types (INTEGER, FLOAT, TIMESTAMP, STRING).

2. **Removing Duplicates:** Duplicate sleep records were identified and removed using `SELECT DISTINCT`.

3. Standardizing Date Formats: Text-based dates were converted to DATE/DATETIME formats to ensure accurate grouping and filtering.

4. Checking for Missing Values: Sparse fields (e.g., weight and fat percentage) were identified.

- Weight data was excluded from primary analysis due to insufficient completeness.

5. Standardizing Column Names: Column names were normalized to [snake_case] to improve readability and joinability.

6. Verifying User ID Consistency: Cross-table checks ensured user IDs aligned across daily, sleep, hourly, and minute-level datasets.

7. Merging Key Tables: Daily activity and sleep data were merged to create a unified daily summary table used for trend analysis.
8. Documenting All Cleaning Steps: All transformations—including duplicate removal, date conversions, missing-value checks, column standardization, table merging, and integrity checks—were documented for reproducibility.

PROCESS Stage — Final Deliverable

By using **BigQuery SQL,** I cleaned and prepared the Fitbit dataset by removing duplicates, standardizing date formats, checking for missing values, and ensuring consistent column naming across all tables. I verified data integrity by checking for mismatched user IDs and incomplete records. Finally, I merged daily, hourly, minute-level, and sleep datasets into a unified semantic layer (M3) that is accurate, consistent, and ready for meaningful trend analysis.

ANALYZE — Explore & Identify Trends

Purpose of This Stage

To explore the cleaned Fitbit dataset using the M3 semantic-layer tables, calculate summary statistics, identify behavioral trends, and connect these insights directly to Bellabeat's business questions. All analysis was performed using BigQuery SQL on the final M3 tables to ensure accuracy, consistency, and analytical readiness.

Organizing the Data for Analysis

The M3 semantic layer provides structured, analysis-ready tables at multiple granularities:

- **M3_daily_master** → daily activity, steps, distance, calories, activity minutes.
- **M3_hourly_master** → hourly steps, calories, intensity, heart rate.
- **M3_minute_master** → long-format minute-level steps, calories, intensity, METs, HR.
- **M3_minuteWide_long** → wide-format minute-level activity for multi-metric comparisons.
- **M3_sleep_master** → sleep duration, time in bed, sleep efficiency, sleep stages.
- **M3_user_summary** → user-level profile metrics (BMI, weight, fat %, device metadata).

These tables were aggregated and joined where necessary to support trend analysis, correlations, and time-series exploration.

Key Trends Identified Across M3 Tables

Trend 1 — Daily Activity Patterns (M3_daily_master)

- Daily steps averaged **7,000–8,000,** below the recommended **10,000.**
- Most users fell into lightly or moderately active categories, with very few high-intensity minutes.
- Strong step-calorie correlation.
- Weekend activity drop.

Implication for Bellabeat: Users track activity but struggle with consistency and intensity — an opportunity for Bellabeat to promote habit-building features and motivational tools.

Trend 2 — Hourly Movement & Intensity (M3_hourly_master)

- Clear movement windows with long periods of inactivity throughout the day.
- Intensity levels were generally low, except for short bursts of activity.
- Heart-rate peaks aligned with intensity bursts, revealing predictable physiological responses.

Implication for Bellabeat: Bellabeat can highlight hourly insights, stress-tracking, and recovery guidance within its app.

Trend 3 — Minute-Level Behavior (M3_minute_master & M3_minuteWide_long)

- micro-patterns of movement, including short bursts of steps and intensity spikes.
- METs and HR trends reveal workout peaks, recovery periods, and stress-related fluctuations.

Implication for Bellabeat: Minute-level insights support personalized coaching, real-time nudges, and stress-reduction features.

Trend 4 — Sleep Duration & Efficiency (M3_sleep_master)

- Average sleep duration was **~6.9 hours,** below the recommended **7–9 hours.**
- Users frequently spend more time in bed than asleep, indicating restlessness or poor sleep efficiency.
- Inconsistent Sleep patterns across days.

Implication for Bellabeat: Strong opportunity to promote sleep-tracking, bedtime reminders, and mindfulness tools.

Trend 5 — User Wellness Profiles (M3_user_summary)

- Moderate activity.
- Irregular sleep.
- Varying heart-rate ranges.

Weight and fat-percentage data were sparse, limiting deeper physiological analysis.

Implication for Bellabeat: Bellabeat can position itself as a holistic wellness ecosystem rather than a fitness-only tracker.

Cross-Table Relationships Identified

Steps ↔ Calories (Daily + Hourly + Minute-Level):

Strong positive correlation was observed across all granularities. More steps consistently led to higher calorie burn.

Sleep ↔ Activity (Daily + Sleep Tables):

Weak Correlation between total minutes asleep and total steps. Being active did not guarantee better sleep.

Intensity ↔ Heart Rate (Hourly + Minute-Level):

Strong alignment, confirming expected physiological responses.

Surprises Discovered in the Data

- Sleep data was more inconsistent than expected.
- High-intensity minutes were extremely low, even on active days.
- Weight and heart-rate data were sparse and not suitable for broad analysis.
- Dramatic Weekend inactivity more typical fitness-tracker assumptions.
- Minute-level data revealed more variability than daily or hourly summaries suggested.

How These Insights Answer the Business Questions

What trends exist in smart-device usage?

- Moderate activity.
- Low intensity.
- Insufficient sleep.
- Weekday-dominant movement.
- Strong step-calorie correlation.

- Predictable HR responses.

How do these trends apply to Bellabeat customers?

Bellabeat customers likely share similar habits — inconsistent sleep, moderate activity, and low high-intensity exercise.

How can these insights shape Bellabeat's marketing strategy?

Insights support messaging around:

- Sleep improvement.
- Habit consistency.
- Weekend motivation.
- Activity-based calorie tracking.
- Holistic wellness guidance.

ANALYZE Stage — Final Deliverable

Using **BigQuery SQL** and the **M3 semantic-layer tables**, I analyzed daily, hourly, minute-level, and sleep data from 30 Fitbit users. The data reveals that; users are moderately active, inconsistently engaged in high intensity workouts, and often sleep below recommended levels. Show significant drops in activity on weekends. Steps strongly correlate with calories burned, while sleep and activity show weak correlation. These insights suggest opportunities for Bellabeat to promote habit-building, sleep improvement, and holistic wellness guidance — directly informing Bellabeat's marketing strategy.

SHARE — Visualize & Present Findings

Purpose of This Stage

To communicate insights through interactive dashboards built from the M3 semantic layer. The dashboards and charts transform raw trends into an accessible narrative for Bellabeat's stakeholders, highlighting user behavior patterns that directly inform marketing strategy.

What Story Does the Data Tell?

Across all M3 tables, the visualizations reveal five major behavioral trends:

- Smart-device users are **moderately active** but rarely reach high-intensity levels.
- Sleep duration is **below recommended levels,** and efficiency varies widely.
- Steps and calories burned show **a strong positive relationship.**
- Activity drops **significantly on weekends**, indicating inconsistent habits.
- Sleep and activity have **a weak correlation**, reinforcing the need for holistic wellness guidance.

These patterns form the foundation of Bellabeat's marketing opportunities.

Dashboards & Visualizations Created (Google Connected Sheets)

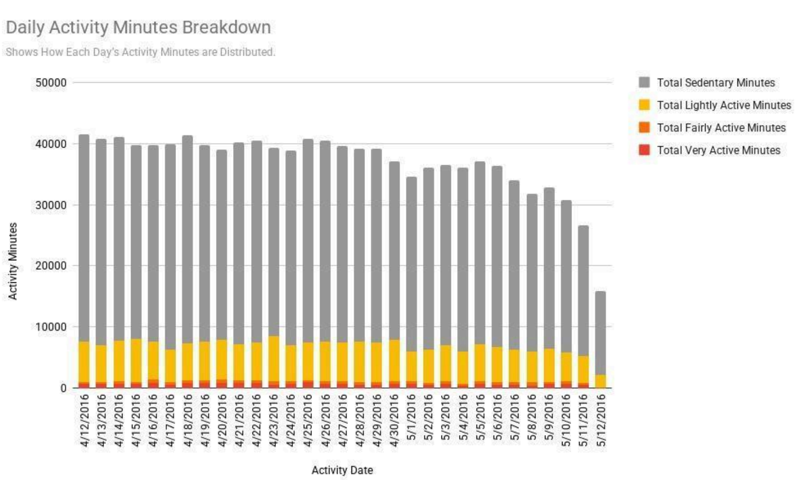
I created **six complete dashboards,** each powered by a dedicated M3 semantic-layer table.

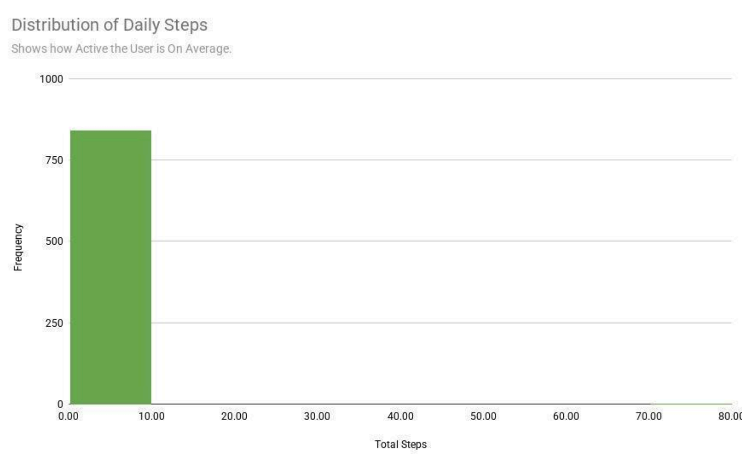
Dashboard 1 — Daily Activity Dashboard (M3_daily_master)

Charts Included:

- Daily Steps Over Time (Line Chart).
- Daily Calories Burned (Line Chart).
- Daily Activity Minutes Breakdown (Stacked Column Chart).
- Steps vs. Calories (Scatter Plot).
- Daily Steps vs. Distance (Scatter Plot).
- Distribution of Daily Steps (Histogram).

Dashboard Focus: Daily trends, activity composition, and daily correlations.



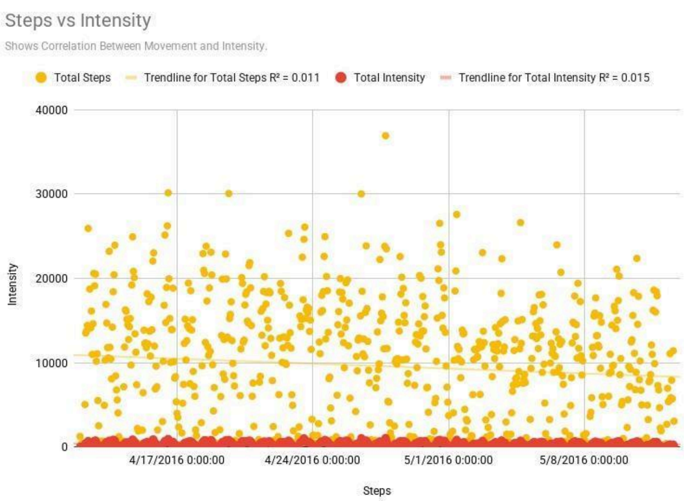


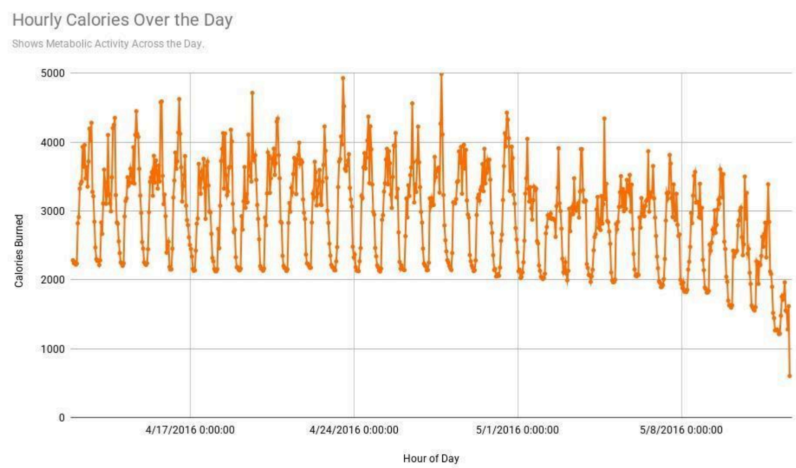
Dashboard 2 — Hourly Activity Dashboard (M3_hourly_master)

Charts Included:

- Hourly Steps Over the Day (Line Chart).
- Hourly Intensity (Heatmap Chart).
- Hourly Calories Over the Day (Line Chart).
- Steps vs. Intensity (Scatter Plot).
- Distribution of Hourly Steps (Histogram).
- Sleep Stages by Hour (Stacked Column Chart).

Dashboard Focus: Hourly movement patterns, intensity windows, and sleep-hour interactions.



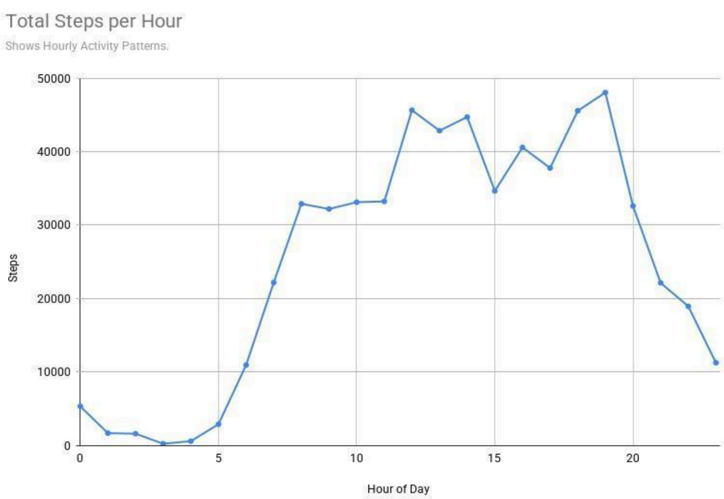


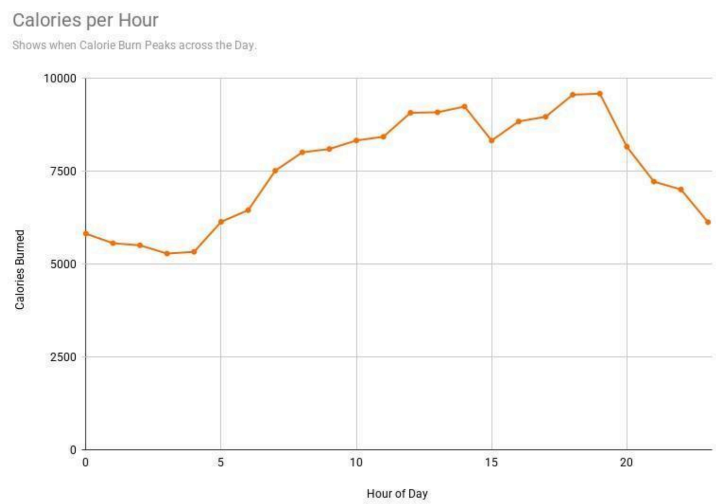
Dashboard 3 — Minute-Wide Dashboard (M3_minuteWide_long)

Charts Included:

- Total Steps per Hour (Line Chart).
- Calories per Hour (Line Chart).
- Intensity per Hour (Line Chart).
- Steps vs. Calories (Scatter Plot).
- Steps Heatmap (Hour × Weekday).
- Steps vs. Intensity (Scatter Plot).
- Intensity Heatmap (Hour × Weekday).

Dashboard Focus: High-resolution activity patterns and weekday/hour behavior cycles.



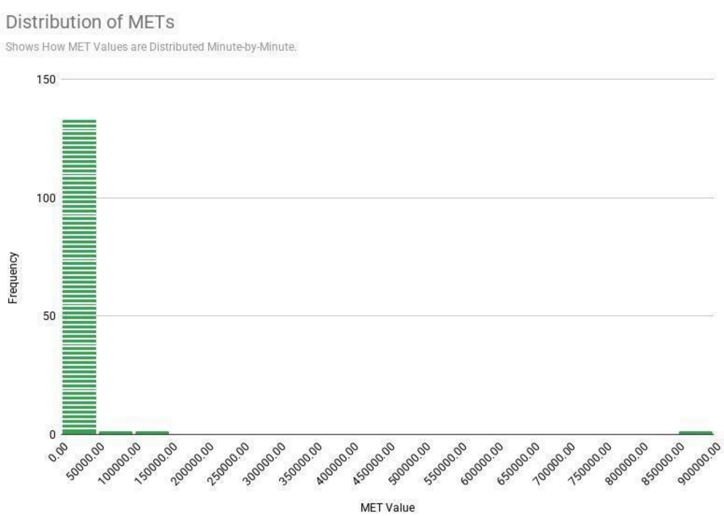


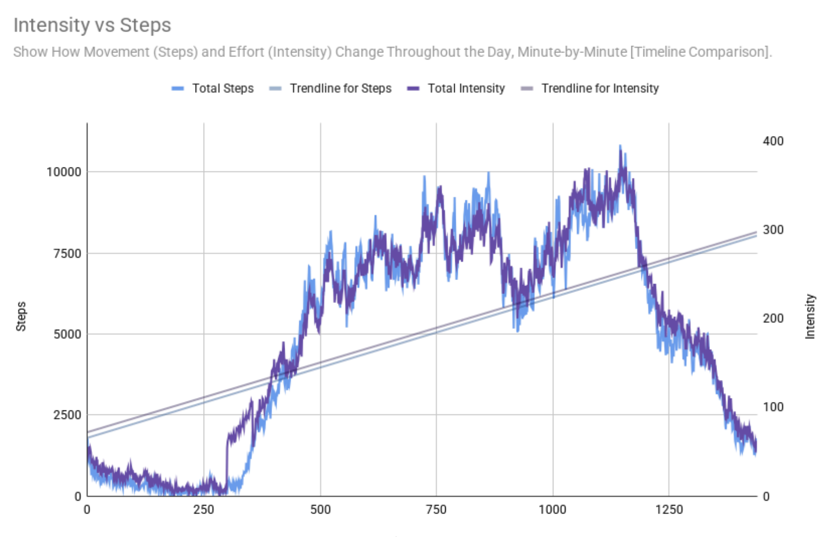
Dashboard 4 — Minute-Level Dashboard (M3_minute_master)

Charts Included:

- Minute-Level Calories Over Time (Line Chart).
- Minute-Level Steps Over Time (Line Chart).
- Intensity vs. Steps (Scatter Plot).
- Intensity vs. Steps (Timeline Comparison Line Chart).
- Distribution of METs (Histogram).
- Sleep Stage Timeline (Stepped Area Chart).
- Hour-Level Steps (Heatmap Chart).

Dashboard Focus: Minute-level intensity, METs distribution, HR patterns, and sleep-stage timelines.



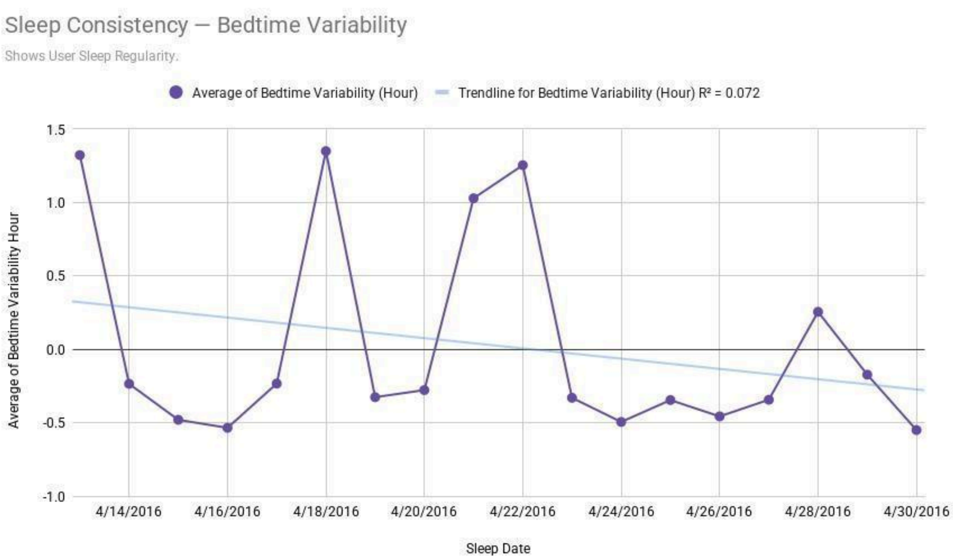


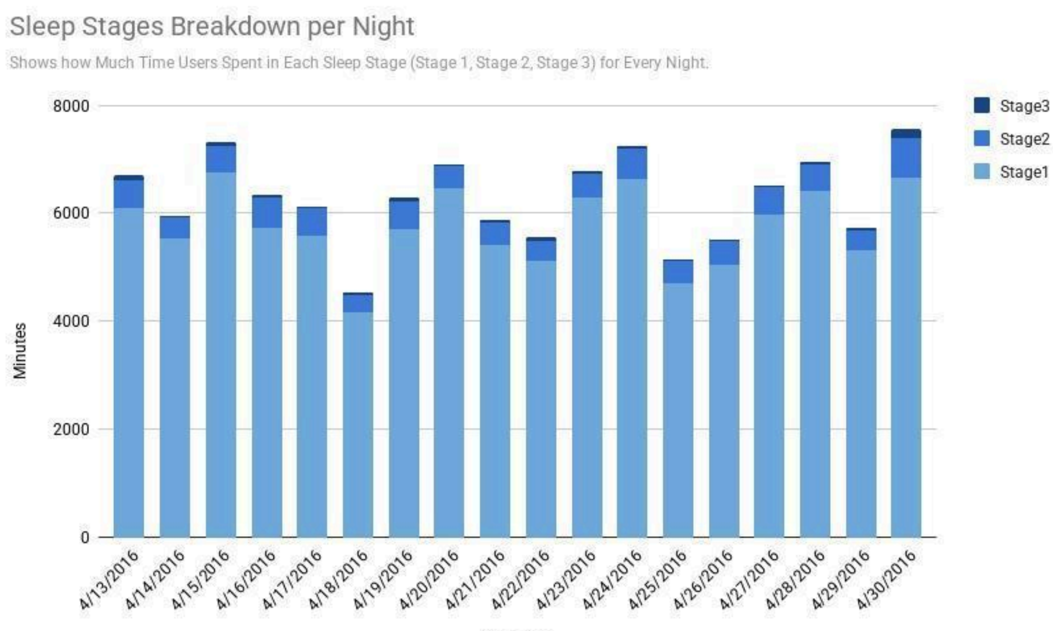
Dashboard 5 — Sleep Dashboard (M3_sleep_master)

Charts Included:

- Sleep Duration per Day (Line Chart).
- Sleep Efficiency per Day (Line Chart).
- Sleep Stages Breakdown per Night (Stacked Bar Chart).
- Bedtime vs. Wake Time (Scatter Plot).
- Sleep Consistency (Bedtime Variability Line Chart).
- Sleep Quality Score Trend (Line Chart).
- Sleep Stage Percentages Over Time (Line Chart).

Dashboard Focus: Sleep duration, efficiency, consistency, and stage composition.



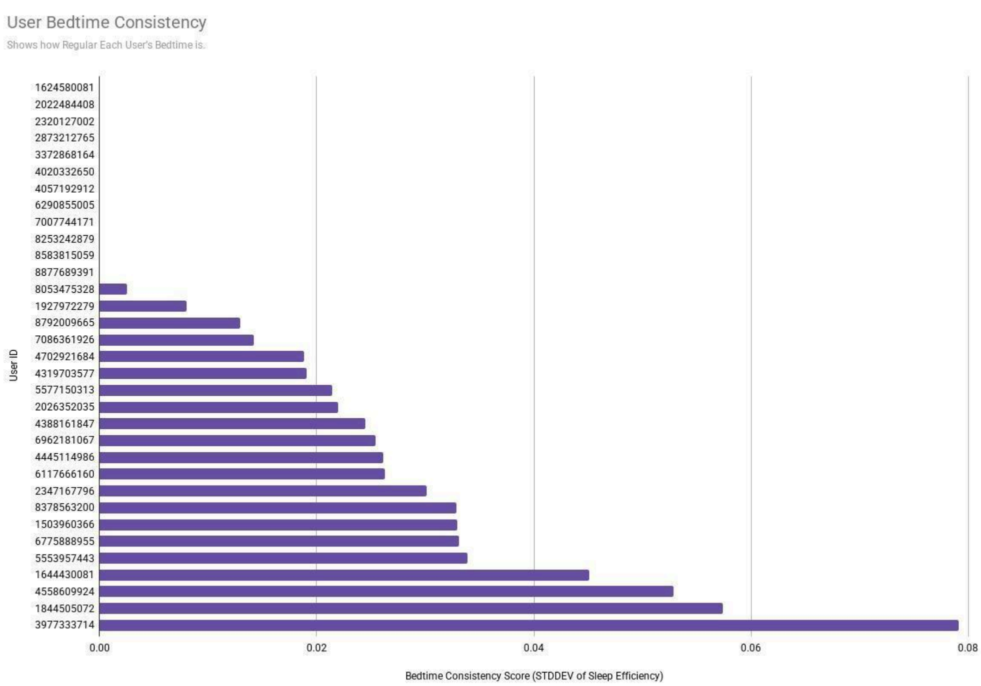


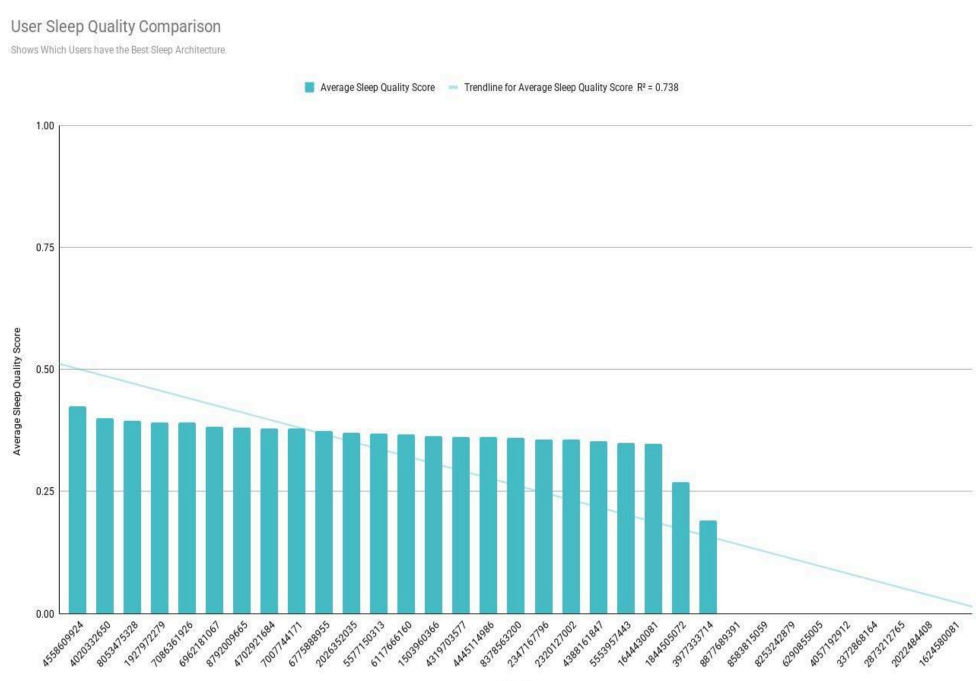
Dashboard 6 — User Summary Dashboard (M3_user_summary)

Charts Included:

- User Sleep Profile (Radar Chart).
- User Sleep Duration Comparison (Bar Chart).
- User Sleep Quality Comparison (Column Chart).
- User Bedtime Consistency (Bar Chart).

Dashboard Focus: Holistic user-level wellness profiles and sleep behavior comparisons.





Key Findings to Present

- Users average **fewer than 10,000 steps per day,** indicating moderate activity levels.
- Sleep duration averages **6.9 hours,** suggesting users struggle with rest and recovery.
- **Strong step-calorie burned correlation**, reinforcing the value of activity tracking.
- Activity **drops significantly** on weekends, showing opportunities for habit-building campaigns.
- **Weak sleep-activity correlation**, supporting Bellabeat's holistic wellness approach.

These insights directly support Bellabeat's marketing strategy for the **Bellabeat App,** Leaf, Time, and **Bellabeat Membership Program.**

SHARE Stage — Final Deliverable

Using **Google Connected Sheets,** I created six dashboards powered by the M3 semantic-layer tables. These dashboards reveal clear patterns in activity, sleep, intensity, and user wellness profiles. Together, they show that users are moderately active, often under-sleep, and exhibit inconsistent wellness habits. These insights will guide Bellabeat's marketing strategy by emphasizing habit-building, sleep improvement, and holistic wellness support.

ACT — Make Recommendations

Purpose of This Stage

To convert insights into actionable marketing strategies; that Bellabeat can use to strengthen product positioning, user engagement, and overall wellness strategy.

These recommendations directly reflect the behavioral trends identified across the **M3_daily_master, M3_hourly_master, M3_minute_master, M3_minuteWide_long, M3_sleep_master, and M3_user_summary** tables.

Top Three Marketing Recommendations for Bellabeat

1. Promote Habit-Building Features to Increase Daily Activity

Insight Behind This Recommendation

Across daily, hourly, and minute-level dashboards, users averaged **7,000–8,000 steps per day** and showed **a significant drop in weekend activity.** High-intensity minutes were consistently low.

Recommendation

Bellabeat should emphasize habit-building tools within the Bellabeat App, Leaf, and Time:

- Daily step goals.
- Weekend activity reminders.
- Personalized movement nudges.
- Streak tracking and achievement badges.

These features encourage consistent movement and help users overcome weekend inactivity.

Why This Matters: Users want motivation and accountability. Bellabeat can position itself as a wellness partner, not just a fitness tracker.

2. Highlight Sleep-Tracking and Wellness Insights to Address Poor Sleep Patterns

Insight Behind This Recommendation

The sleep dashboard revealed that users average **~6.9 hours of sleep,** often spend more time in bed than asleep, and show irregular patterns. Sleep efficiency fluctuates significantly.

Recommendation

Bellabeat should market its sleep-tracking capabilities more aggressively, especially through the Bellabeat App and Time watch:

- Sleep duration and quality insights.
- Bedtime reminders.
- Stress-reduction and mindfulness tools.
- Personalized sleep improvement plans.

Why This Matters: Sleep is a major wellness concern. Bellabeat's female-focused brand is uniquely positioned to address stress, restlessness, and lifestyle-related sleep issues.

3. Position Bellabeat as a Holistic Wellness Ecosystem, Not Just a Fitness Tracker

Insight Behind This Recommendation

Across **M3_user_summary** and cross-table correlations, **sleep and activity were weakly correlated**, indicating users need whole-person wellness support — not just step tracking.

Recommendation

Bellabeat should emphasize its holistic ecosystem:

- Bellabeat Membership Program.
- Nutrition guidance.
- Mindfulness and stress-tracking.
- Hydration tracking with Spring.
- Women's cycle and reproductive health insights.

Marketing should highlight how Bellabeat integrates **activity, sleep, stress, hydration, and women's health** into one seamless experience.

Why This Matters: This differentiates Bellabeat from competitors like Fitbit and Apple Watch, which focus heavily on fitness metrics rather than holistic wellness.

ACT Stage — Final Deliverable

Based on the analysis of user behavior across all M3 semantic-layer dashboards, I recommend that Bellabeat focus on:

1) **Promoting habit-building features** to increase daily activity and reduce weekend inactivity,

2) **Highlighting sleep-tracking and wellness insights** to address insufficient sleep patterns, and

3) **Positioning Bellabeat as a holistic wellness ecosystem** that supports women's overall health.

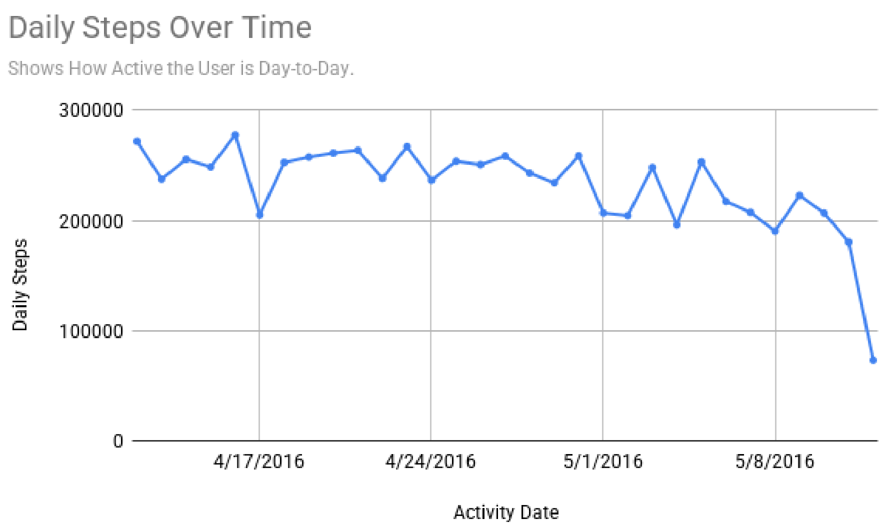
These strategies align with observed user behavior trends and will help Bellabeat strengthen engagement, differentiate its products, and support.

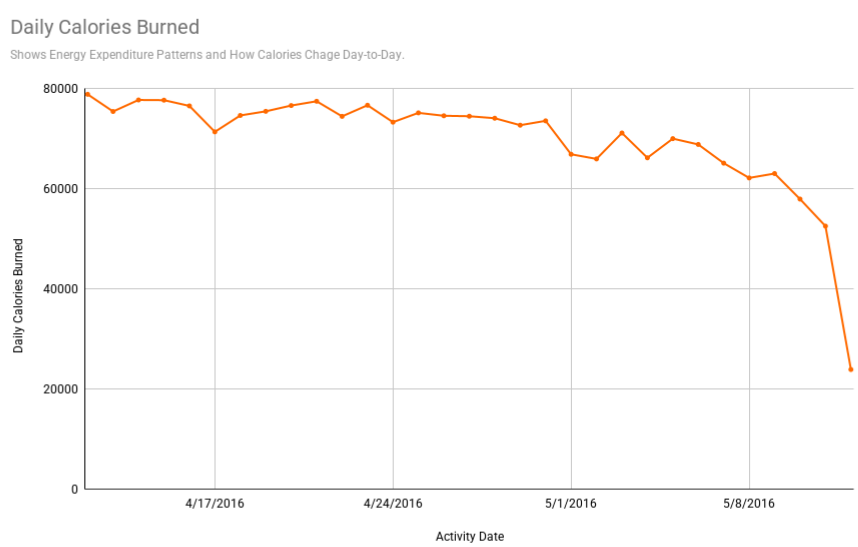
APPENDIX

APPENDIX A – Dashboards & Charts

This appendix contains all charts used in the SHARE stage. Each dashboard is presented with its full set of charts for reference. Charts are displayed in color, while all text remains in black.

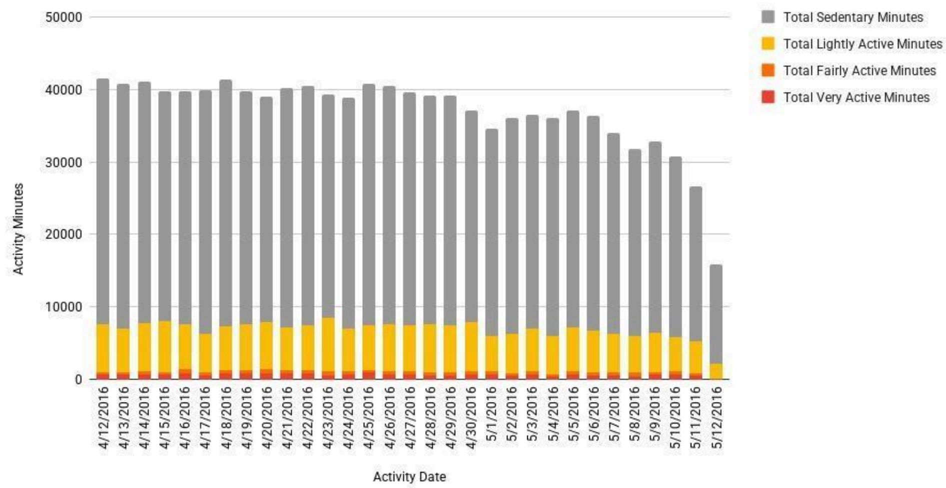
Dashboard 1 — Daily Activity Dashboard (M3_daily_master)





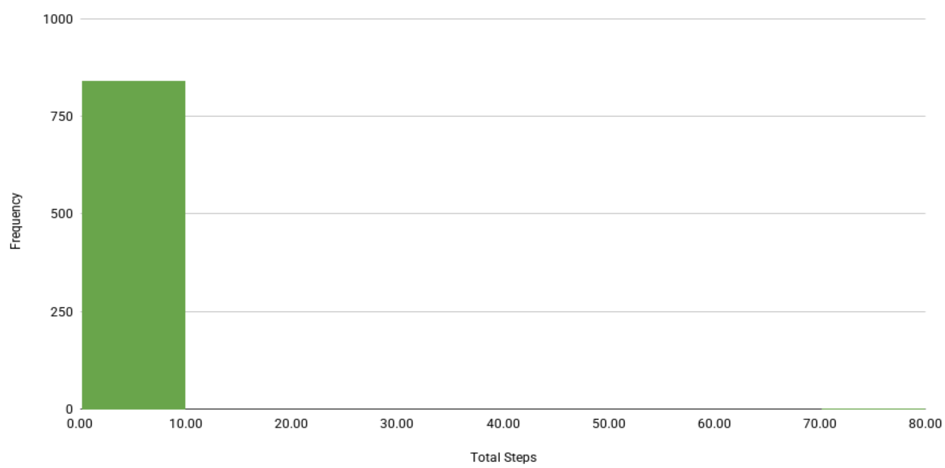
Daily Activity Minutes Breakdown

Shows How Each Day's Activity Minutes are Distributed.

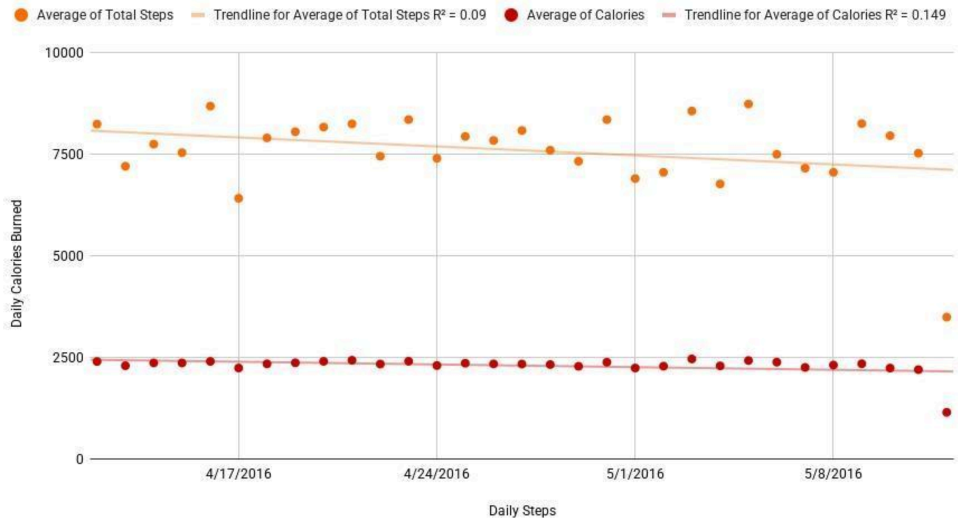


Distribution of Daily Steps

Shows how Active the User is On Average.

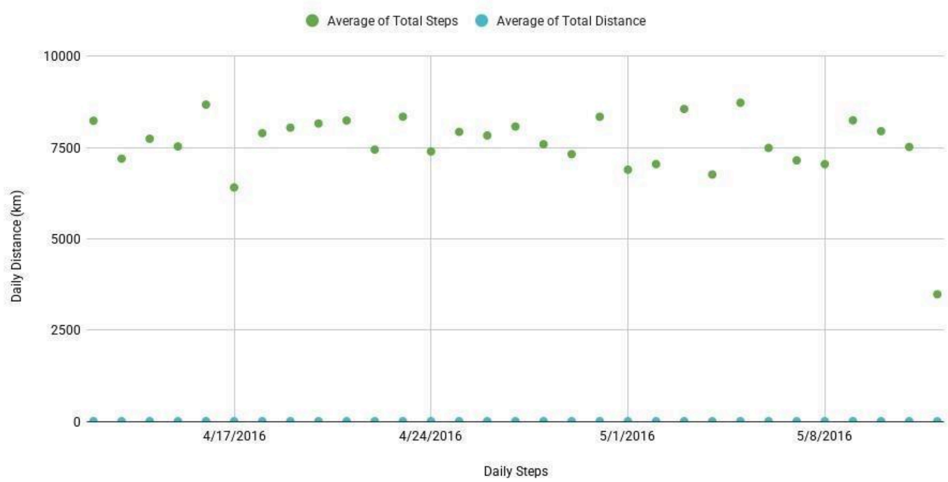


Daily Steps vs Calories

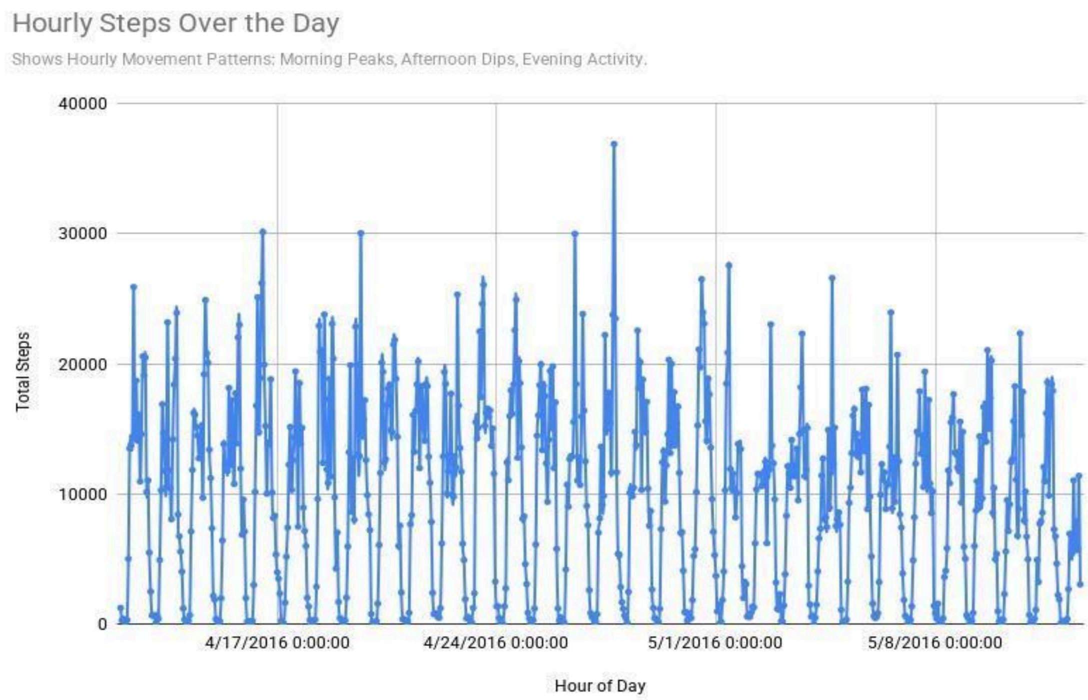


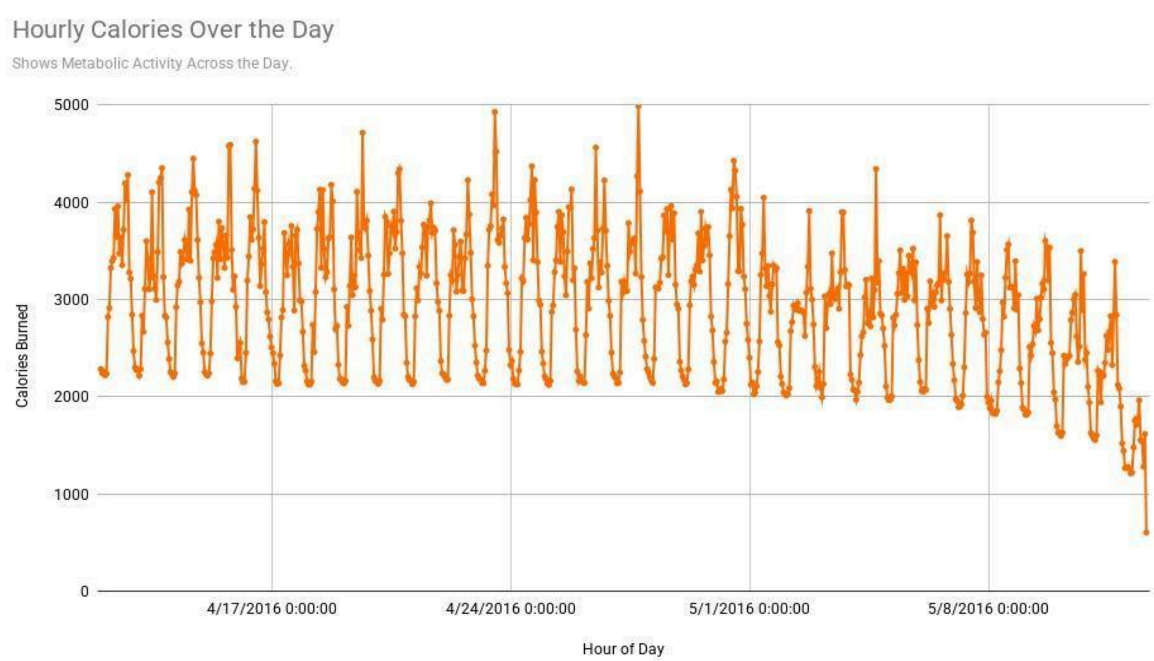
Daily Steps vs Distance

Shows the Relationship between Total Steps Taken and Total Distance Traveled per Day.



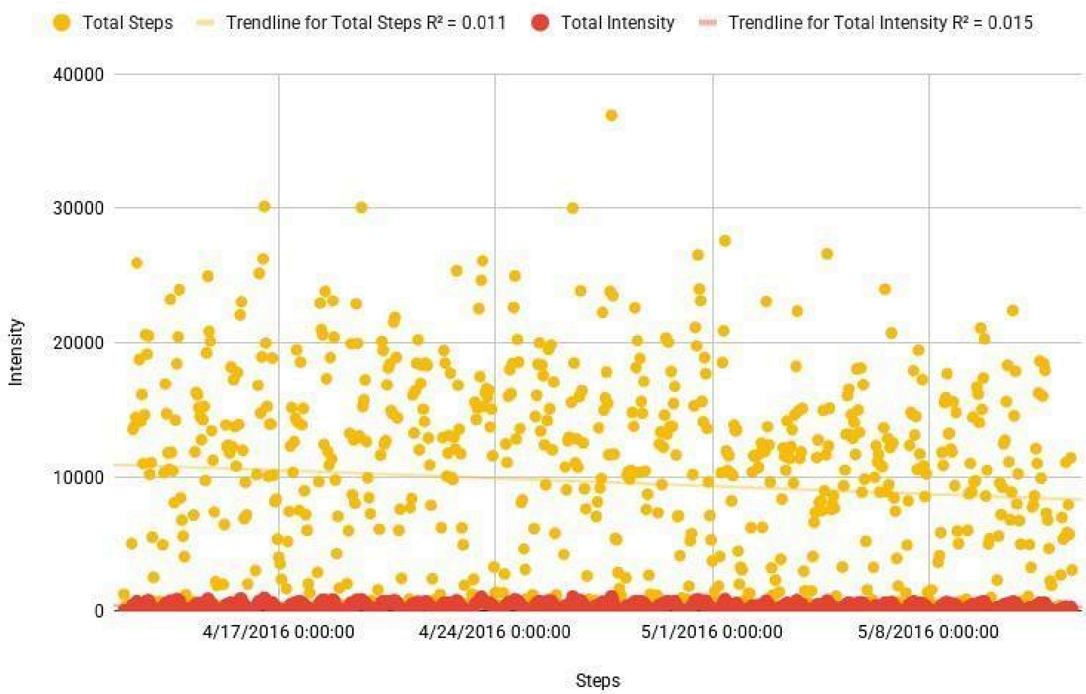
Dashboard 2 — Hourly Activity Dashboard (M3_hourly_master)





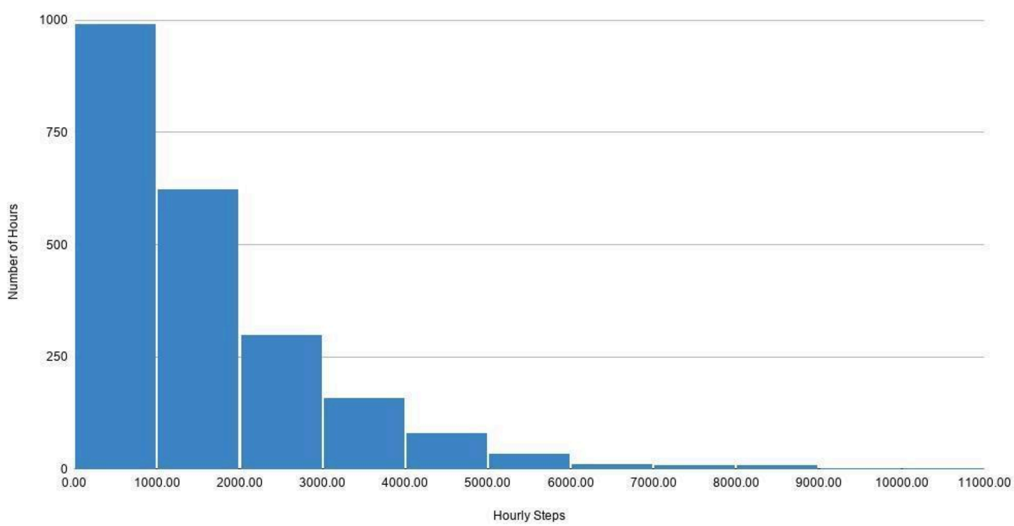
Steps vs Intensity

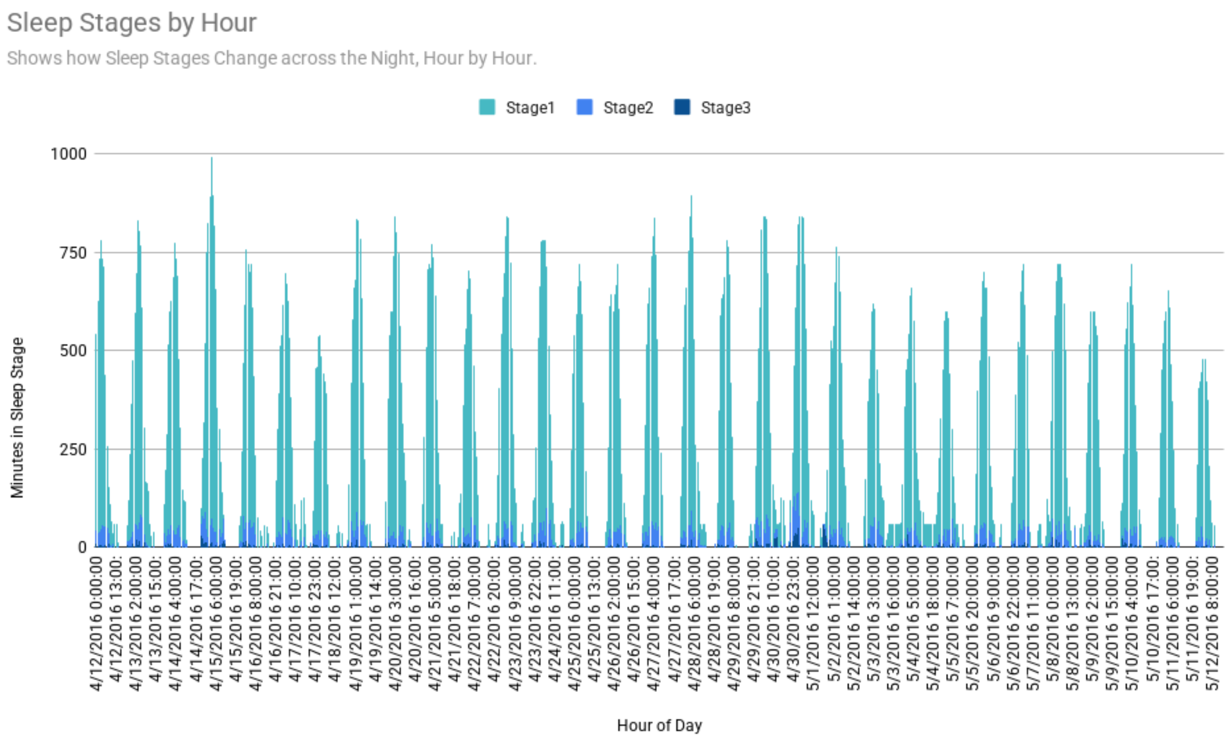
Shows Correlation Between Movement and Intensity.



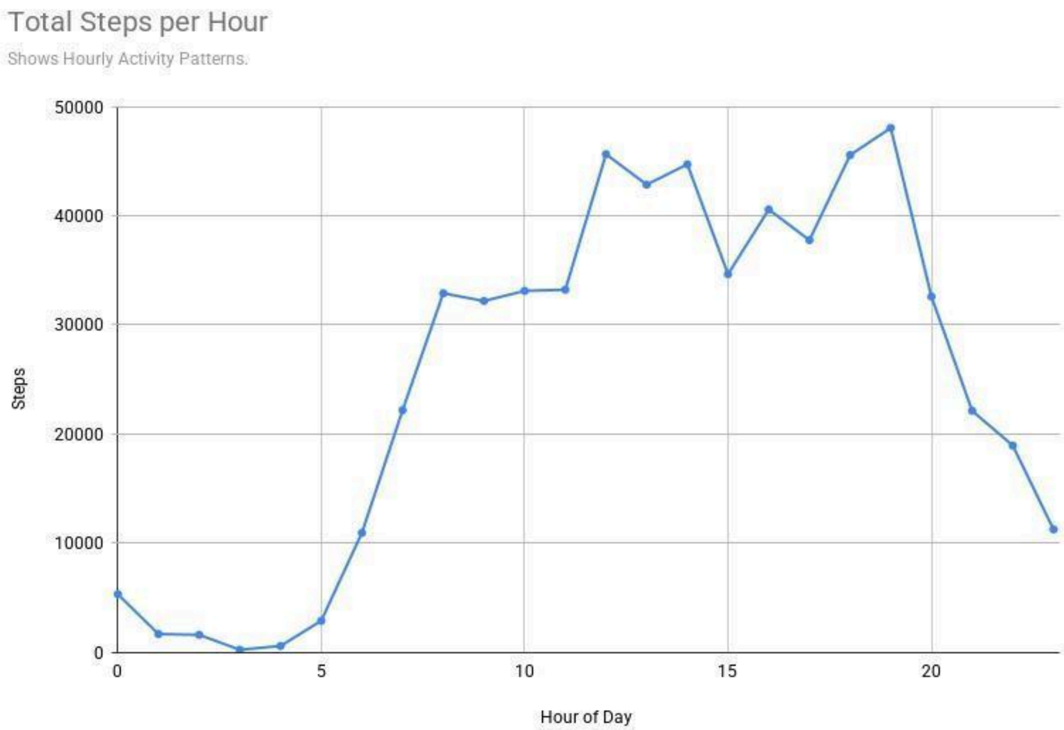
Distribution of Hourly Steps

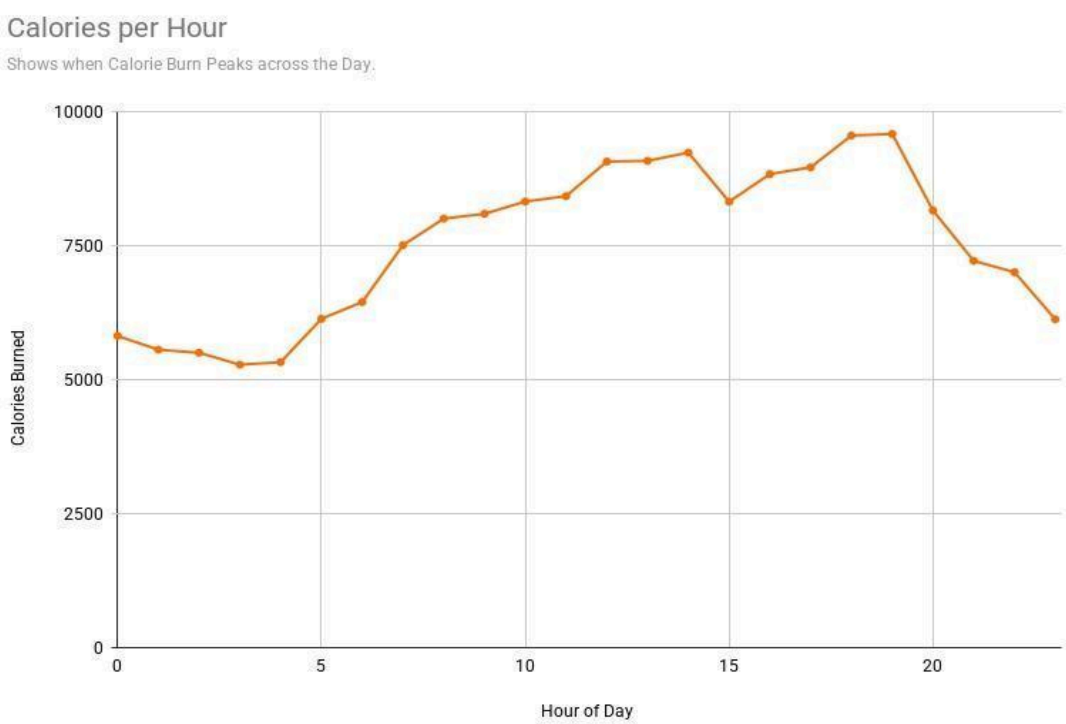
Shows How Steps are Distributed Across Hours.




Sleep Stages by Hour
Shows how Sleep Stages Change across the Night, Hour by Hour.
Stage1
Stage2
Stage3
1000
750
500
250
0
Minutes in Sleep Stage
4/12/2016 0:00:00
4/12/2016 13:00:
4/13/2016 2:00:00
4/13/2016 15:00:
4/14/2016 4:00:00
4/14/2016 17:00:
4/15/2016 6:00:00
4/15/2016 19:00:
4/16/2016 8:00:00
4/16/2016 21:00:
4/17/2016 10:00:
4/17/2016 23:00:
4/18/2016 12:00:
4/19/2016 1:00:00
4/19/2016 14:00:
4/20/2016 3:00:00
4/20/2016 16:00:
4/21/2016 5:00:00
4/21/2016 18:00:
4/22/2016 7:00:00
4/22/2016 20:00:
4/23/2016 9:00:00
4/23/2016 22:00:
4/24/2016 11:00:
4/25/2016 0:00:00
4/25/2016 13:00:
4/26/2016 2:00:00
4/26/2016 15:00:
4/27/2016 4:00:00
4/27/2016 17:00:
4/28/2016 6:00:00
4/28/2016 19:00:
4/29/2016 8:00:00
4/29/2016 21:00:
4/30/2016 10:00:
4/30/2016 23:00:
5/1/2016 12:00:00
5/2/2016 1:00:00
5/2/2016 14:00:00
5/3/2016 3:00:00
5/3/2016 16:00:00
5/4/2016 5:00:00
5/4/2016 18:00:00
5/5/2016 7:00:00
5/5/2016 20:00:00
5/6/2016 9:00:00
5/6/2016 22:00:00
5/7/2016 11:00:00
5/8/2016 0:00:00
5/8/2016 13:00:00
5/9/2016 2:00:00
5/9/2016 15:00:00
5/10/2016 4:00:00
5/10/2016 17:00:
5/11/2016 6:00:00
5/11/2016 19:00:
5/12/2016 8:00:00
Hour of Day

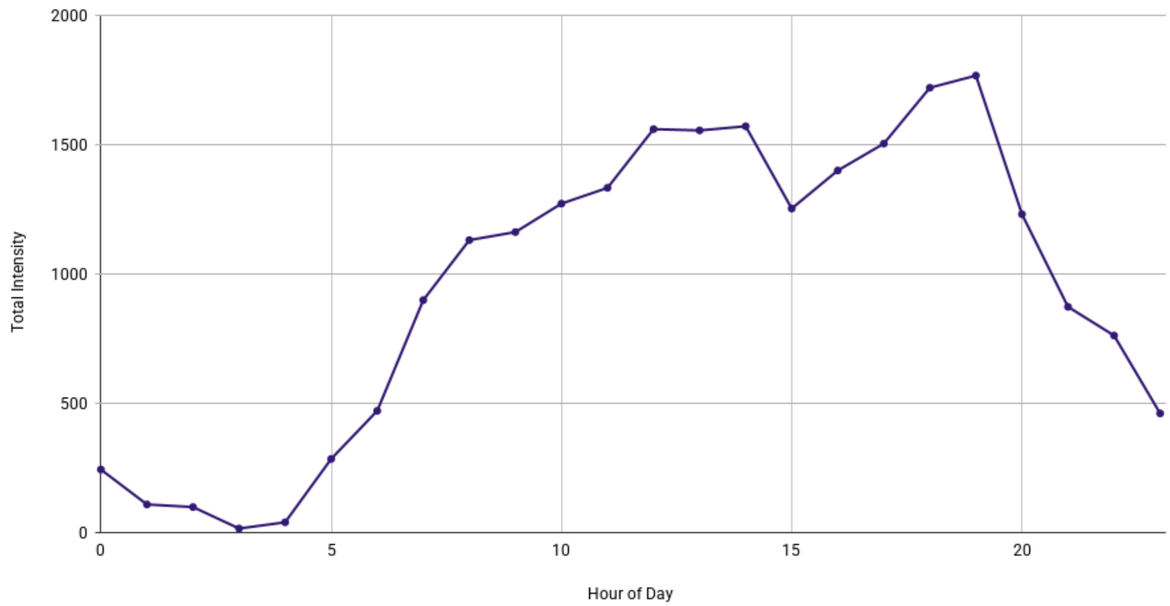
Dashboard 3 — Minute-Wide Dashboard (M3_minuteWide_long)





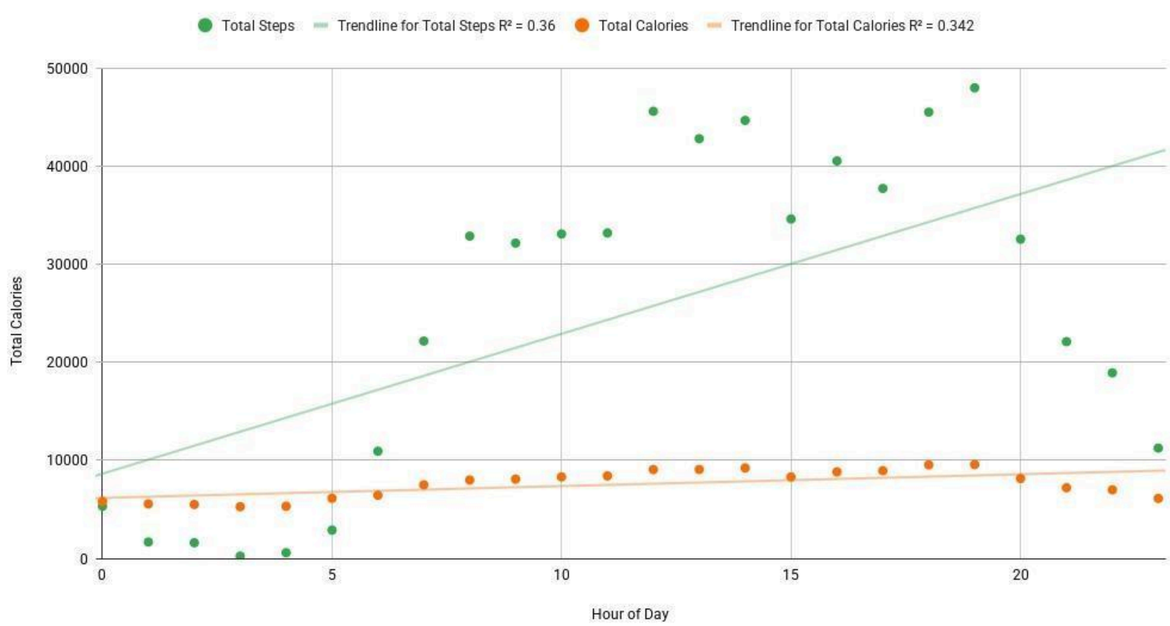
Intensity per Hour

Show Total Intensity Accumulated During Each Hour Across The Day.

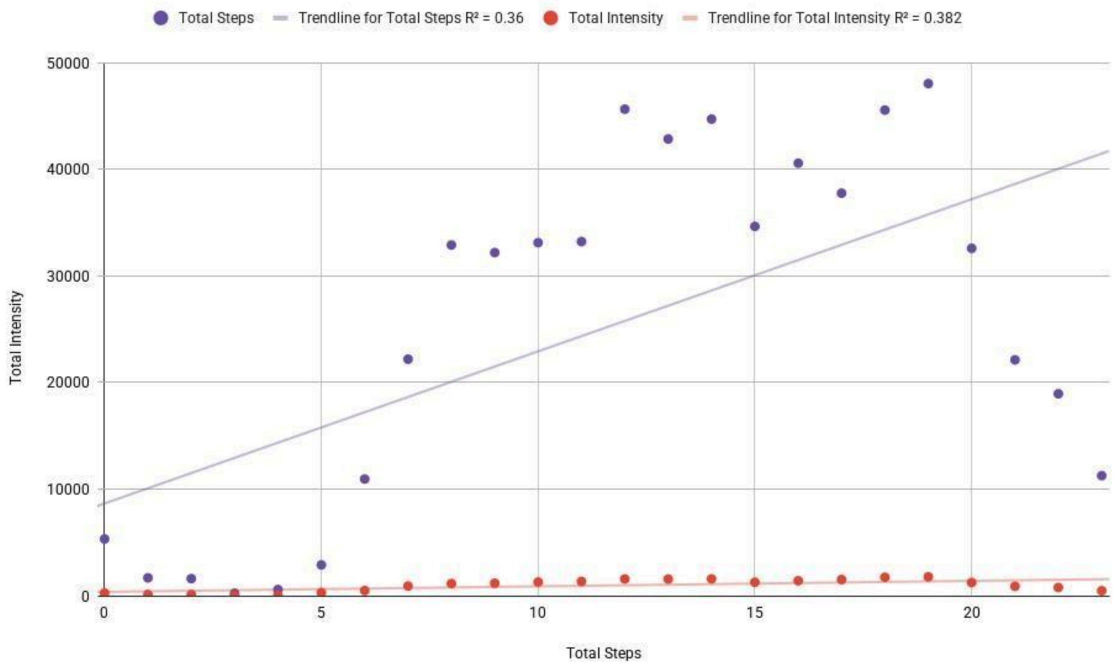


Steps vs Calories

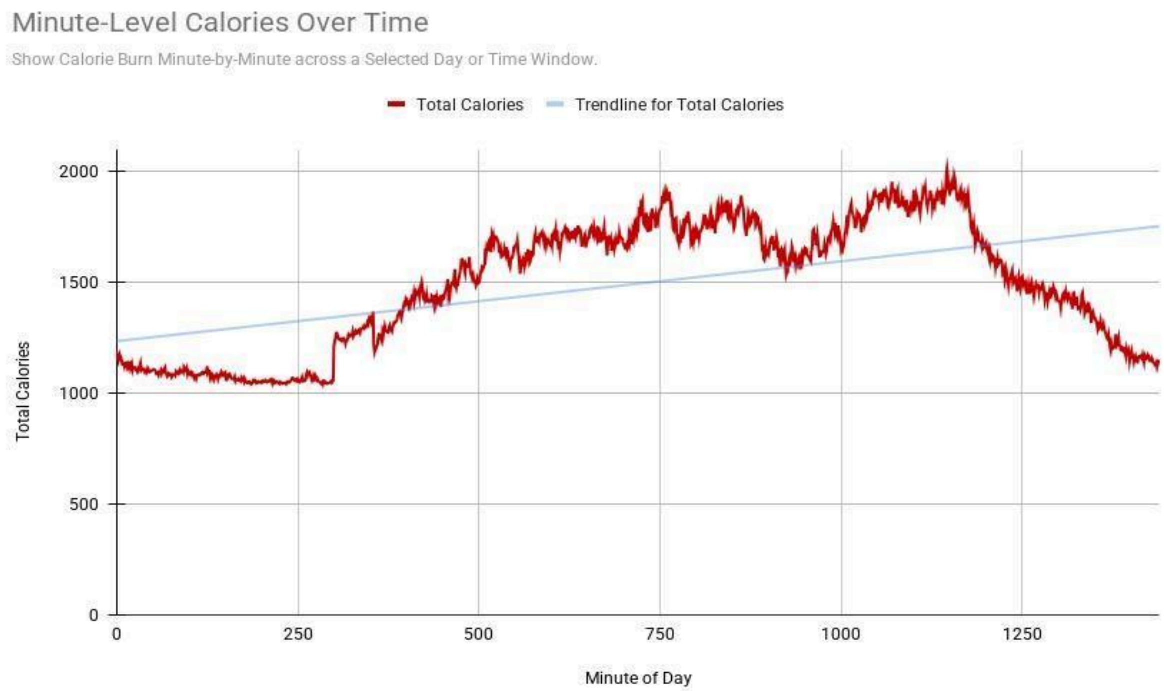
Shows the Relationship Between Total Hourly Steps and Total Hourly Calories Burned.

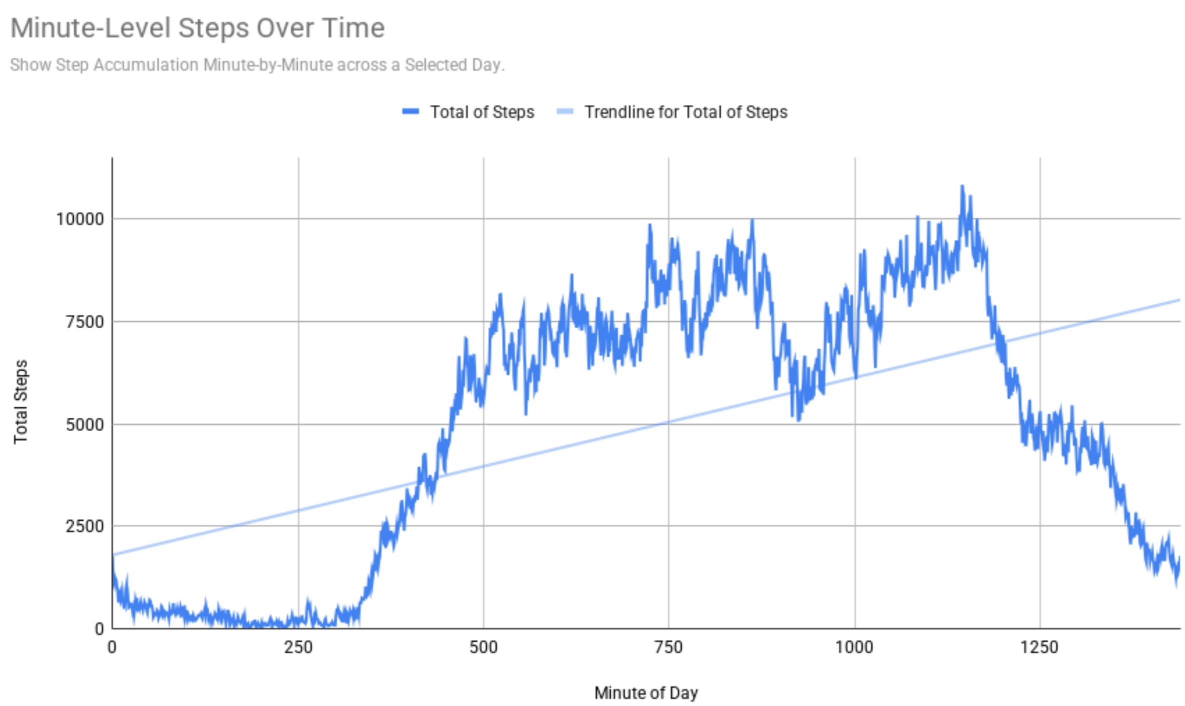


Steps vs Intensity



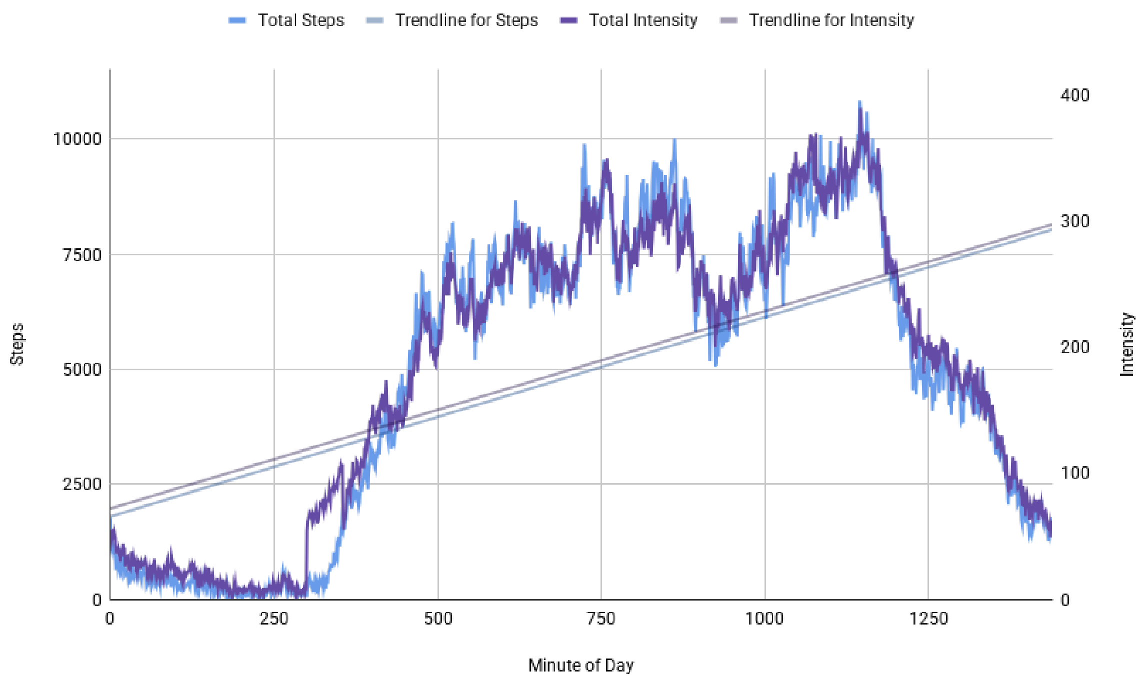
Dashboard 4 — Minute-Level Dashboard (M3_minute_master)





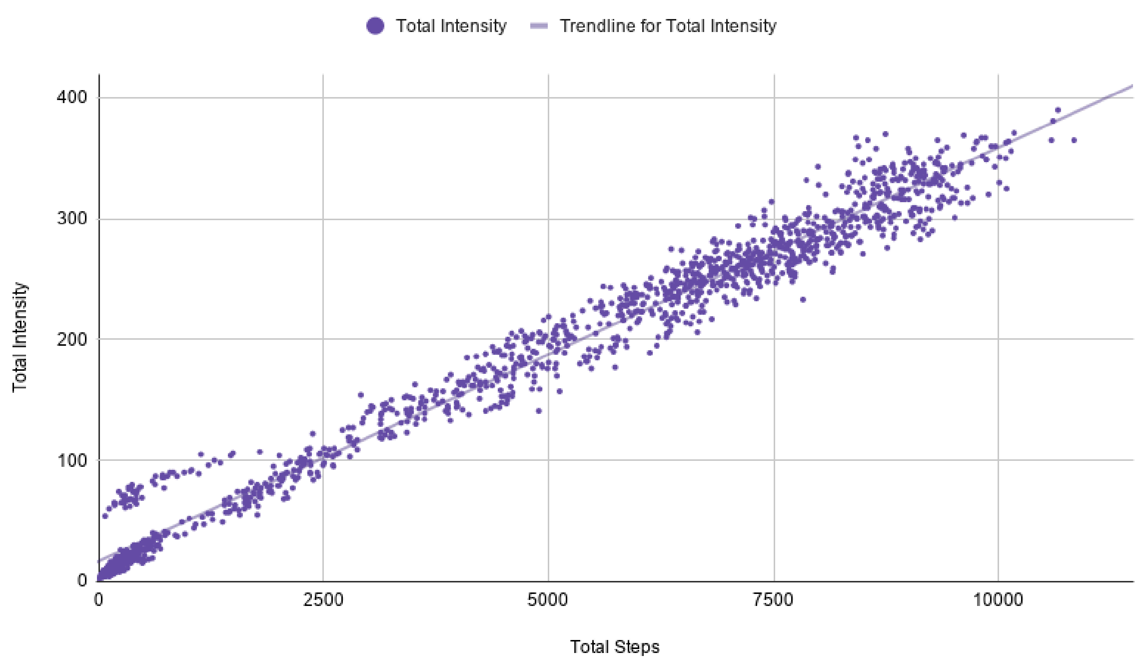
Intensity vs Steps

Show How Movement (Steps) and Effort (Intensity) Change Throughout the Day, Minute-by-Minute [Timeline Comparison].



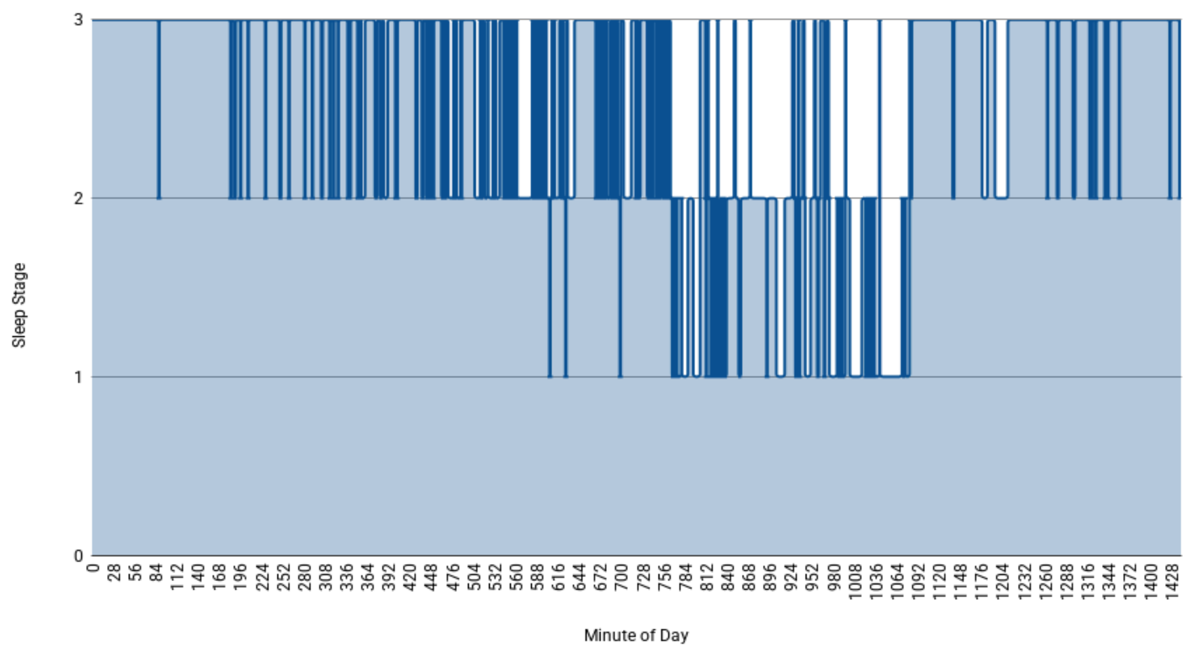
Intensity vs Steps

Shows the Relationship between Movement (Steps) and Effort (Intensity) at the Minute Level.



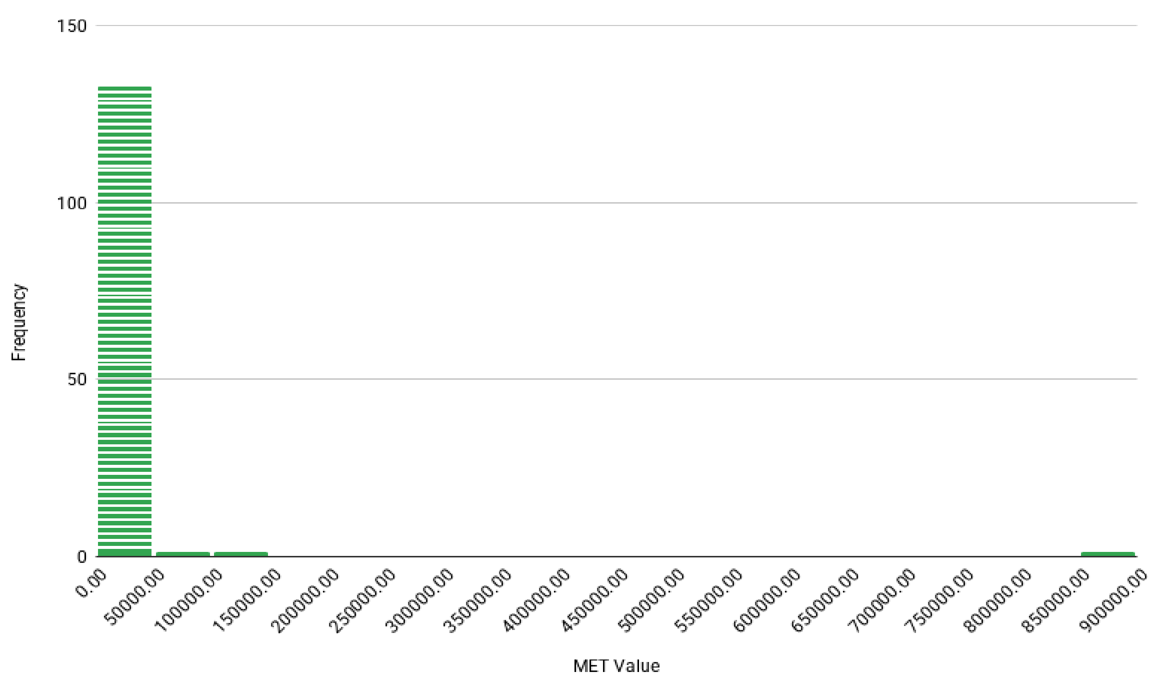
Sleep Stage Timeline

Show Sleep Stages Over Time.

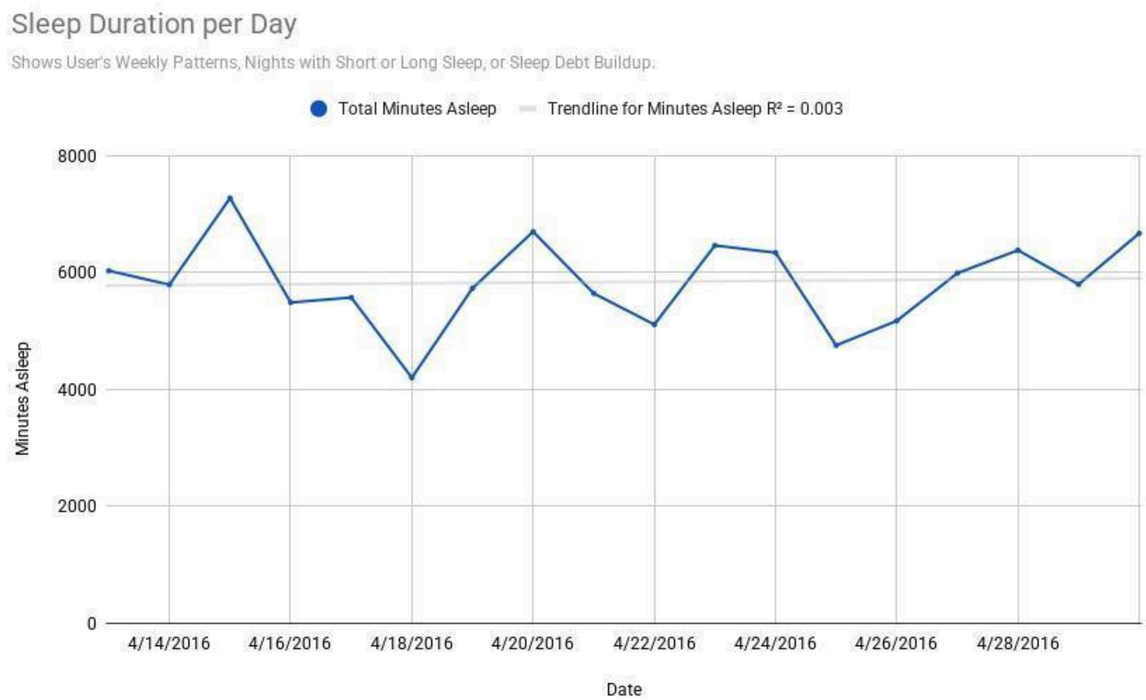


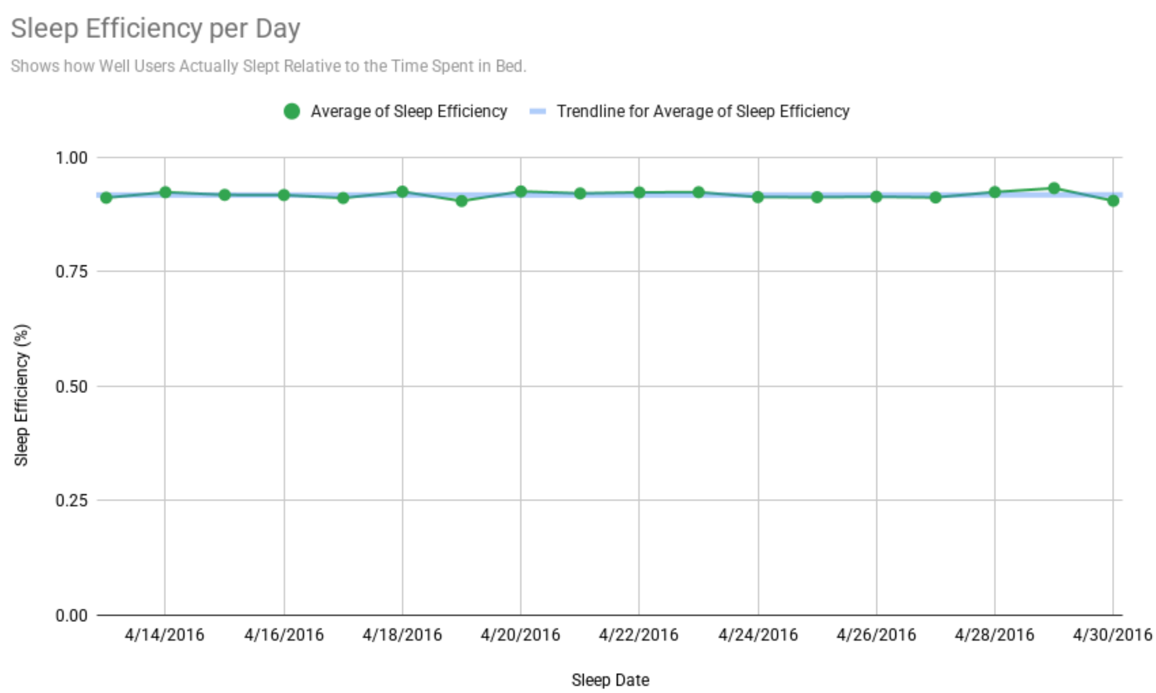
Distribution of METs

Shows How MET Values are Distributed Minute-by-Minute.



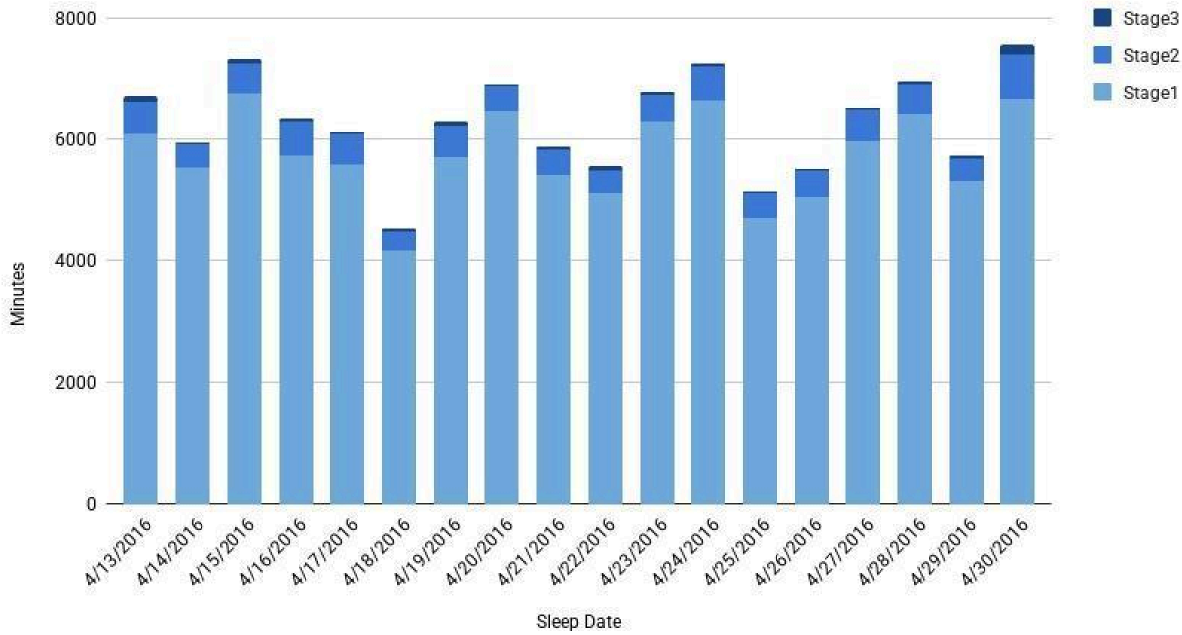
Dashboard 5 — Sleep Dashboard (M3_sleep_master)





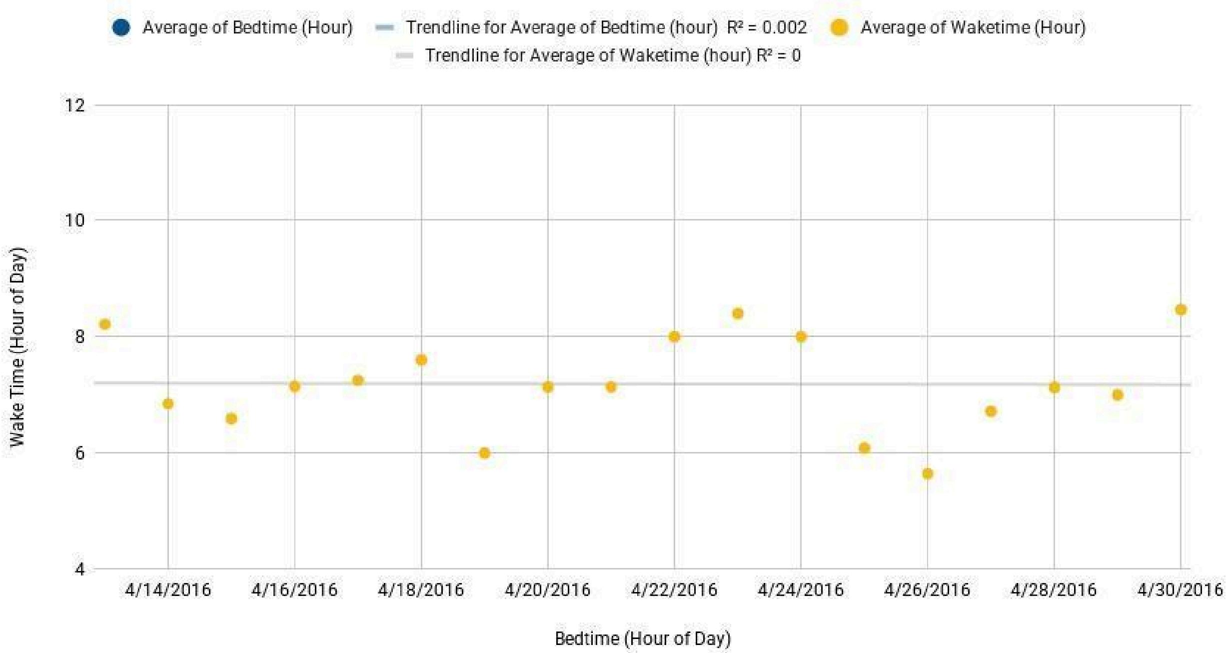
Sleep Stages Breakdown per Night

Shows how Much Time Users Spent in Each Sleep Stage (Stage 1, Stage 2, Stage 3) for Every Night.



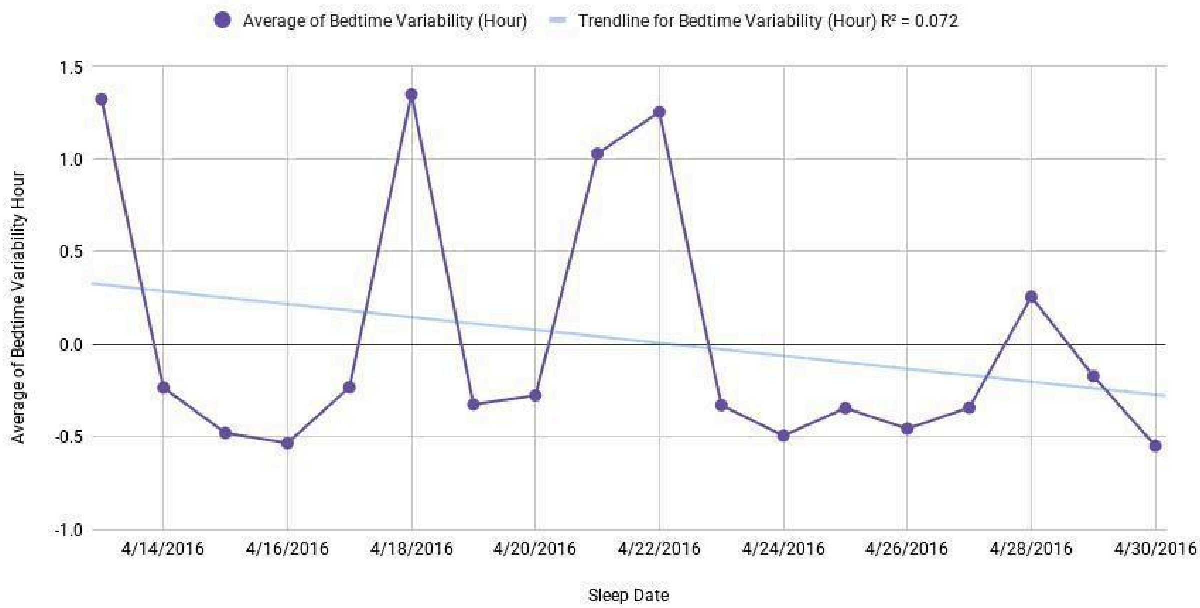
Bedtime vs Wake Time (True Sleep Session)

Shows When Users go to Bed Wake Up, Revealing Consistency, Drift, and Weekend Patterns.



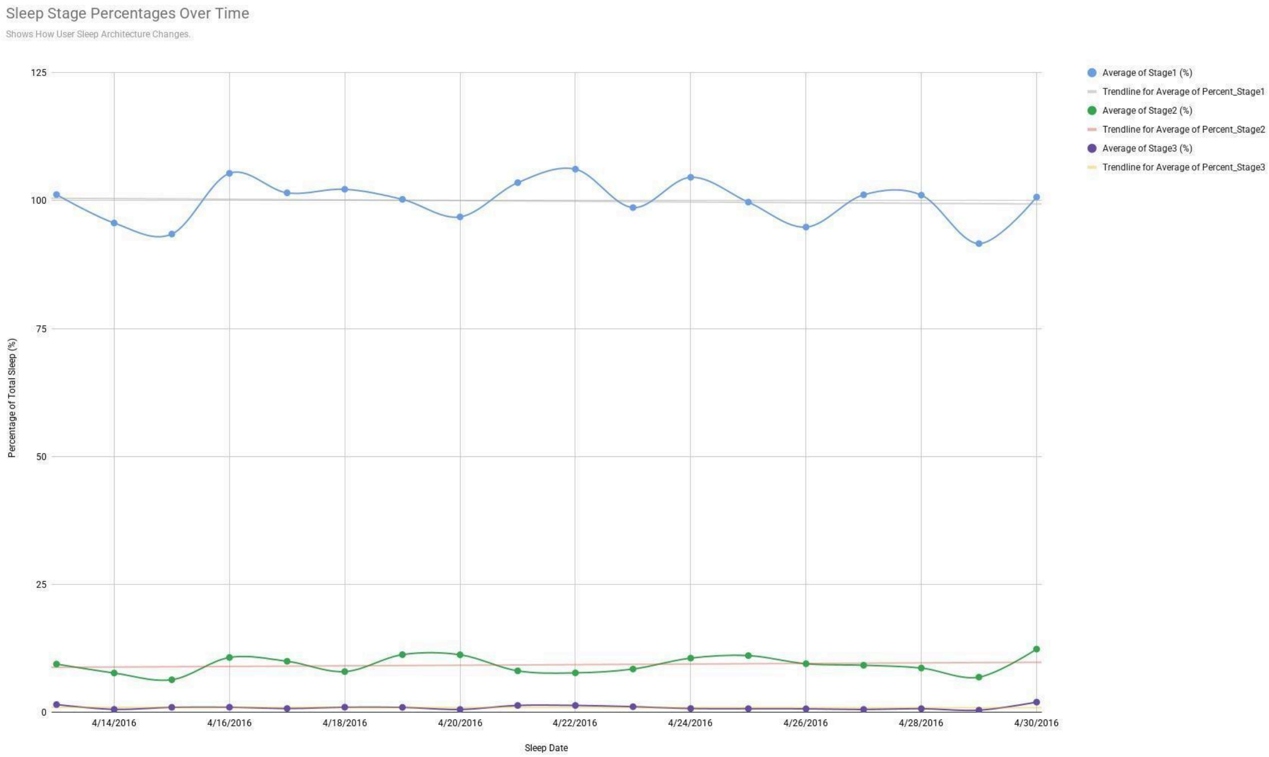
Sleep Consistency — Bedtime Variability

Shows User Sleep Regularity.



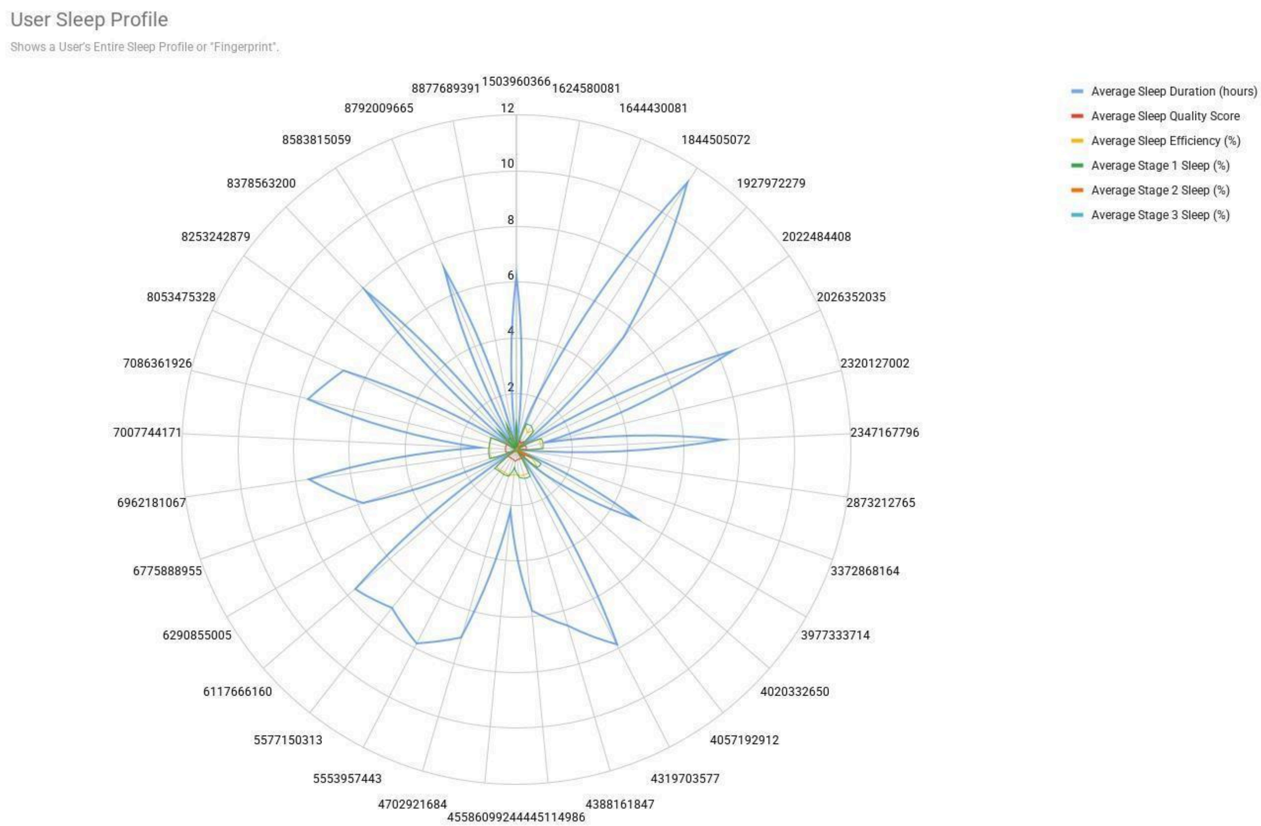
Sleep Quality Score Trend

Shows How User Sleep Quality Changes Over Time.



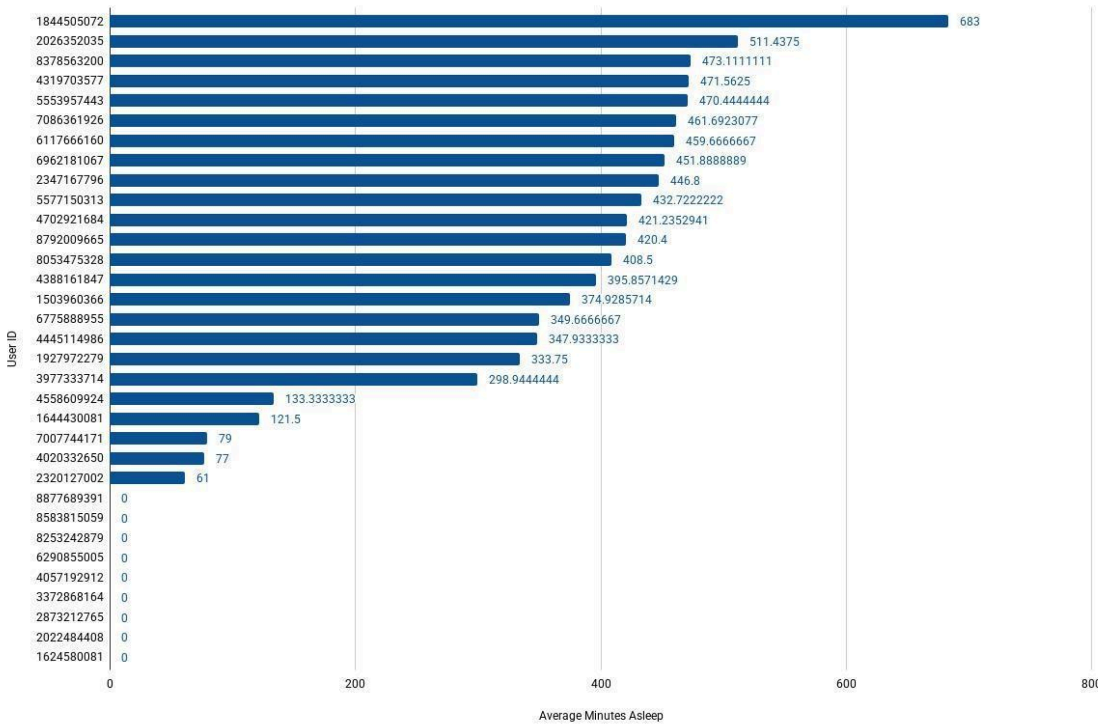
Sleep Stage Percentages Over Time
Shows How User Sleep Architecture Changes.
125
100
75
50
25
0
Percentage of Total Sleep (%)
4/14/2016
4/16/2016
4/18/2016
4/20/2016
4/22/2016
4/24/2016
4/26/2016
4/28/2016
4/30/2016
Sleep Date
Average of Stage1 (%)
Trendline for Average of Percent_Stage1
Average of Stage2 (%)
Trendline for Average of Percent_Stage2
Average of Stage3 (%)
Trendline for Average of Percent_Stage3

Dashboard 6 — User Summary Dashboard (M3_user_summary)



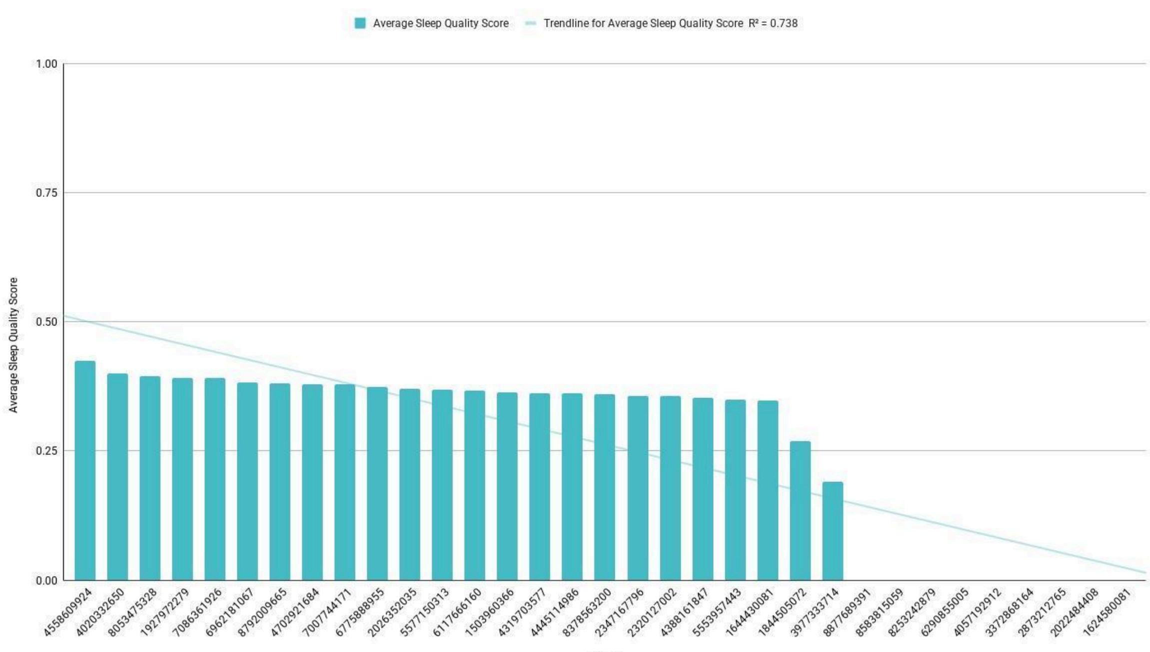
User Sleep Duration Comparison

Shows Which Users Sleep More or Less on Average.



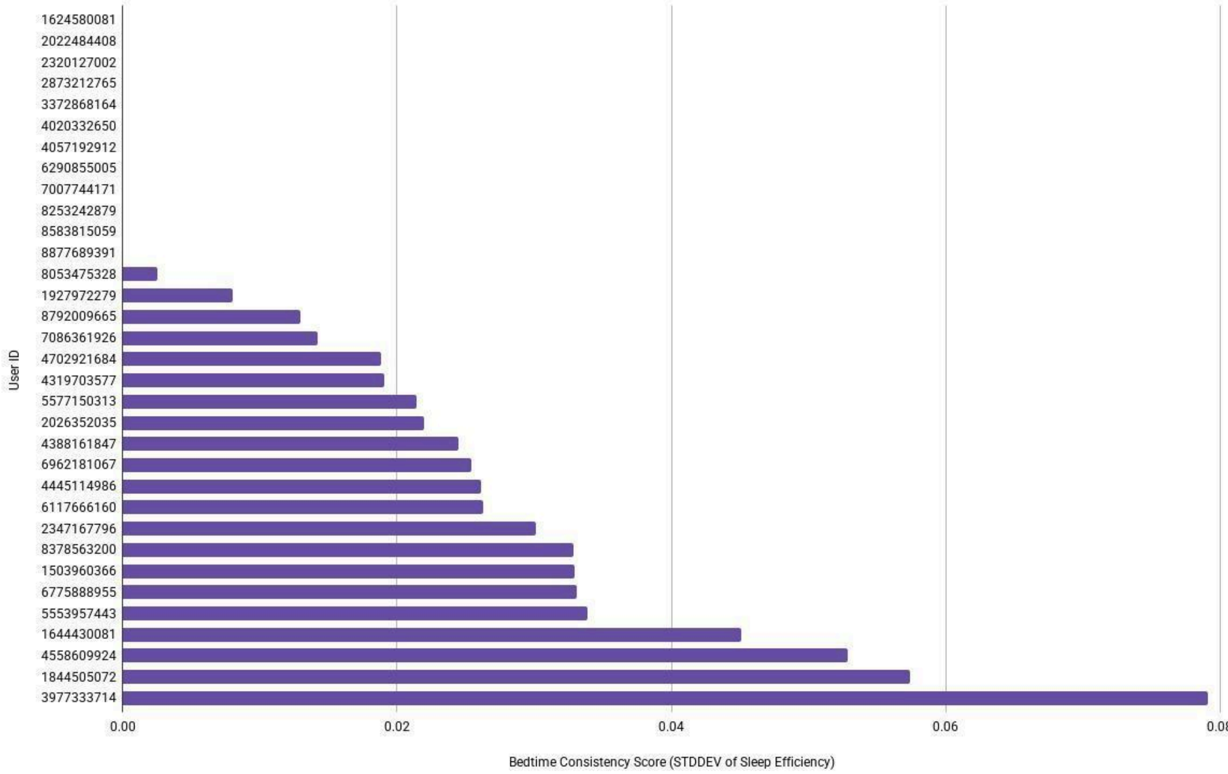
User Sleep Quality Comparison

Shows Which Users have the Best Sleep Architecture.



User Bedtime Consistency

Shows how Regular Each User's Bedtime is.



APPENDIX B — Data Dictionary

This section defines all fields used in the M3 semantic layer tables. The below tables follow a consistent 3-column structure.

B1. M3_daily_master

The M3_daily_master table contains daily-level activity, distance, sleep, heart-rate, and METs metrics. It provides a high-level view of each user's daily wellness patterns and aggregates data from minute-level and hourly sources.

Field Names	Description	Data Type
user_id	Unique user identifier	INTEGER
activity_date	Date of recorded activity	DATE
total_steps	Total steps recorded for the day	INTEGER
total_distance	Total distance traveled **(km)**	FLOAT
tracker_distance	Distance measured by device tracker	FLOAT
logged_activities_distance	Distance from logged activities	FLOAT
very_active_distance	Distance during very active movement	FLOAT
moderately_active_distance	Distance during moderate activity	FLOAT
light_active_distance	Distance during light activity	FLOAT
sedentary_active_distance	Distance during sedentary movement	FLOAT
very_active_minutes	Minutes of very active movement	INTEGER
fairly_active_minutes	Minutes of fairly active movement	INTEGER
lightly_active_minutes	Minutes of light activity	INTEGER
sedentary_minutes	Minutes spent sedentary	INTEGER
calories_activity	Calories burned from activity	INTEGER

calories_calories	Total calories burned **(all sources)**	INTEGER
total_sleep_records	Number of sleep records for the day	INTEGER
total_minutes_asleep	Total minutes asleep	INTEGER
total_time_in_bed	Total minutes spent in bed	INTEGER
weight_kg	Weight in kilograms	FLOAT
bmi	Body mass index	FLOAT
fat_percent	Body fat percentage	INTEGER
is_manual_report	Whether weight/fat data was manually reported	BOOLEAN
avg_hr	Average heart rate for the day	FLOAT
min_hr	Minimum heart rate recorded	INTEGER
max_hr	Maximum heart rate recorded	INTEGER
total_calories_minute	Sum of minute-level calories	FLOAT
total_steps_minute	Sum of minute-level steps	INTEGER
total_intensity_minute	Sum of minute-level intensity values	INTEGER
total_mets_minute	Sum of minute-level METs values	INTEGER
minutes_stage1	Total minutes in Stage 1 sleep	INTEGER
minutes_stage2	Total minutes in Stage 2 sleep	INTEGER
minutes_stage3	Total minutes in Stage 3 **(Deep)** sleep	INTEGER

B2. M3_hourly_master

The M3_hourly_master table contains hourly-level activity, intensity, calorie burn, METs, and sleep-stage metrics. It supports mid-granularity trend analysis between daily and minute-level datasets.

Field Name	Description	Data Type
user_id	Unique user identifier	INTEGER
activity_hour	Hour timestamp **(YYYY-MM-DD HH:00:00)**	TIMESTAMP
activity_date	Date extracted from the hour timestamp	DATE
calories	Total calories burned during the hour	INTEGER
total_intensity	Total intensity score for the hour	INTEGER
average_intensity	Average intensity score for the hour	INTEGER
steps	Total steps recorded during the hour	INTEGER
avg_hr	Average heart rate during the hour	FLOAT
min_hr	Minimum heart rate recorded in the hour	INTEGER
max_hr	Maximum heart rate recorded in the hour	INTEGER
total_calories_minute	Sum of minute-level calories within the hour	FLOAT
total_steps_minute	Sum of minute-level steps within the hour	INTEGER
total_intensity_minute	Sum of minute-level intensity values	INTEGER
total_mets_minute	Sum of minute-level METs values	INTEGER
minutes_stage1	Total minutes in Stage 1 sleep during the hour	INTEGER
minutes_stage2	Total minutes in Stage 2 sleep during the hour	INTEGER
minutes_stage3	Total minutes in Stage 3 **(Deep)** sleep during the hour	INTEGER

B3. M3_minute_master

The M3_minute_master table contains long-format minute-level activity metrics including steps, calories, intensity, METs, heart-rate values, and sleep-stage indicators. It supports granular time-series analysis across the entire day.

Field Name	Description	Data Type
user id	Unique user identifier	INTEGER
activity_minute	Exact minute timestamp **(YYYY-MM-DD HH:MM:00)**	TIMESTAMP
activity_date	Date extracted from the minute timestamp	DATE
calories	Calories burned during the minute	FLOAT
steps	Steps recorded during the minute	INTEGER
intensity	Intensity score **(0–3 scale)**	INTEGER
mets	Metabolic equivalent value	INTEGER
avg_hr	Average heart rate during the minute	FLOAT
min_hr	Minimum heart rate recorded in the minute	INTEGER
max_hr	Maximum heart rate recorded in the minute	INTEGER
sleep_stage	Sleep stage code **(0=Awake, 1=Light, 2=Deep, 3=REM)**	INTEGER
minute_of_day	Minute index within the day **(0–1439)**	INTEGER

B4. M3_minuteWide_long

The M3_minuteWide_long table contains wide-format minute-level activity metrics, allowing multi-metric comparisons across steps, calories, intensity, and temporal attributes such as hour and weekday.

Field Name	Description	Data Type
user_id	Unique user identifier	INTEGER
activity_hour	Hour label for grouping **(e.g., “2020-05-01 14:00”)**	STRING
minute_index	Minute index within the hour **(0–59)**	INTEGER
calories	Calories burned during the minute	FLOAT
intensity	Intensity score **(0–3 scale)**	INTEGER

steps	Steps recorded during the minute	INTEGER
hour_of_day	Hour of day **(0–23)**	INTEGER
weekday_sort	Numeric weekday order **(1=Mon … 7=Sun)**	INTEGER
weekday_label	Weekday name **(Mon–Sun)**	STRING

B5. M3_sleep_master

The M3_sleep_master table contains nightly sleep session metrics including timestamps, sleep stages, efficiency, and variability indicators. It provides detailed insight into user sleep behavior and quality.

Field Name	**Description**	**Data Type**
user_id	Unique user identifier	INTEGER
sleep_date	Date of the sleep session	DATE
sleep_start_timestamp	Timestamp when sleep began	TIMESTAMP
sleep_end_timestamp	Timestamp when sleep ended	TIMESTAMP
true_bedtime	Actual bedtime timestamp	TIMESTAMP
true_waketime	Actual wake-up timestamp	TIMESTAMP
bedtime_hour	Hour of bedtime **(0–23)**	INTEGER
waketime_hour	Hour of wake-up **(0–23)**	INTEGER
total_minutes_asleep	Total minutes asleep	INTEGER
total_time_in_bed	Total minutes spent in bed	INTEGER
total_sleep_records	Number of sleep records contributing to a session	INTEGER
minutes_stage1	Minutes in Stage 1 sleep	INTEGER
minutes_stage2	Minutes in Stage 2 sleep	INTEGER
minutes_stage3	Minutes in Stage 3 **(Deep)** sleep	INTEGER

percent_stage1	Percentage of Stage 1 sleep	FLOAT
percent_stage2	Percentage of Stage 2 sleep	FLOAT
percent_stage3	Percentage of Stage 3 sleep	FLOAT
sleep_efficiency	Sleep efficiency **(%)**	FLOAT
user_bedtime_mean_hour	Average bedtime hour across sessions	FLOAT
bedtime_variability_hour	Variability of bedtime hour	FLOAT
sleep_quality_score	Composite sleep quality score	FLOAT

B6. M3_user_summary

The M3_user_summary table contains aggregated user-level wellness metrics derived from daily, hourly, minute-level, minute-wide, and sleep datasets. It provides a holistic profile of each user's activity, sleep, and intensity patterns.

Field Name	Description	Data Type
user_id	Unique user identifier	INTEGER
days_tracked	Total number of days with valid data	INTEGER
avg_daily_steps	Average daily steps	FLOAT
avg_daily_distance	Average daily distance **(km)**	FLOAT
avg_daily_calories	Average daily calories burned	FLOAT
avg_very_active_minutes	Average daily very active minutes	FLOAT
avg_fairly_active_minutes	Average daily fairly active minutes	FLOAT
avg_lightly_active_minutes	Average daily lightly active minutes	FLOAT
avg_sedentary_minutes	Average daily sedentary minutes	FLOAT
sleep_sessions	Total number of sleep sessions recorded	INTEGER
avg_minutes_asleep	Average minutes asleep per session	FLOAT

avg_time_in_bed	Average time in bed per session	FLOAT
avg_sleep_efficiency	Average sleep efficiency **(%)**	FLOAT
avg_stage1_pct	Average % of Stage 1 sleep	FLOAT
avg_stage2_pct	Average % of Stage 2 sleep	FLOAT
avg_stage3_pct	Average % of Stage 3 **(Deep)** sleep	FLOAT
avg_hourly_steps	Average steps per hour	FLOAT
avg_hourly_calories	Average calories burned per hour	FLOAT
avg_hourly_intensity	Average intensity per hour	FLOAT
avg_minute_steps	Average steps per minute	FLOAT
avg_minute_calories	Average calories per minute	FLOAT
Avg_minute_intensity	Average intensity per minute	FLOAT
total_minuteWide_records	Total minute-wide records available	INTEGER
avg_minuteWide_steps	Average steps in **minute-wide format**	FLOAT
avg_minuteWide_calories	Average calories in **minute-wide format**	FLOAT
avg_minuteWide_intensity	Average intensity in **minute-wide format**	FLOAT
Avg_sleep_duration_hours	Average sleep duration **(hours)**	FLOAT
Avg_sleep_quality_score	Average sleep quality score	FLOAT
Sleep_to_activity_ratio	Ratio of sleep duration to activity levels	FLOAT
Activity_balance_score	Composite score balancing activity intensity and consistency	FLOAT
Per_user_sleep_efficiency_consistency	Variability/consistency of sleep efficiency	FLOAT

APPENDIX C — External Links

Daily Activity Dashboard

https://docs.google.com/spreadsheets/d/1UOqBIzbxtHZRBdhJlFk2UcH5h-UdU_7KVMHd3e2ROcQ/edit?usp=sharing

Hourly Activity Dashboard

https://docs.google.com/spreadsheets/d/1scManRyHREv8tnpX3gGXSY9OhLgiTueRgxRgiwYA82I/edit?usp=sharing

MinuteWide Dashboard

https://docs.google.com/spreadsheets/d/1hOnMj-HXMjLeWlrGffjNKbtzvIsYdjRMN4CfiuE0Wl4/edit?usp=sharing

Minute-Level Dashboard

https://docs.google.com/spreadsheets/d/1tj-XxEVKZuhCtj2IuweE-I8Y02JyQov_W50baYtrIu8/edit?usp=sharing

Sleep Dashboard

https://docs.google.com/spreadsheets/d/1VxxcoE4JY0id6vi1PguhUSfXOHevwvI9jxmcDmJm9nI/edit?usp=sharing

User Summary Dashboard

https://docs.google.com/spreadsheets/d/1IzoubRoQDlBGPcrF5XeXVSnXooWbvTtSLN5s2feZVSw/edit?usp=sharing